

Getting to Know Libero

Course Workbook

Table of Contents

About this Workbook

[Slide 3](#)

Pre-Lab: Workshop Pre-requisites

[Slide 4](#)

Lab 1: Understanding Vivado

[Slide 7](#)

About this Workbook

This workbook is designed to be used in conjunction with the Getting to Know Libero workbook.

The contents of this workbook are created by Aduvo Engineering & Training, Ltd.

If you have any questions about the contents, or need assistance, please contact Adam Taylor at adam@adiuvoengineering.com.

Pre-Lab

Workshop Pre-requisites

Required Hardware

There is no required hardware for this course.

Downloads and Installations

Step 1 – Download and install the following at least one day prior to the workshop. This may take a significant amount of time and drive space.

Libero SoC 11.9	Download
Source Project Files	Download

Lab 1

Overview and Introduction to Libero

Lab 1: Overview and Introduction to Libero

Step 1 – Open Libero 11.9



Lab 1: Overview and Introduction to Libero

Step 2 – Create a new project from the project menu



Lab 1: Overview and Introduction to Libero

Step 3 – Enter a project name and location – select the HDL type as VHDL – leave block creation unchecked

The screenshot shows the 'New project' dialog box in the Libero IDE. The 'Project details' step is selected in the left sidebar. The main area contains the following fields and options:

- Project name:** An empty text input field.
- Project location:** A text input field containing the path `C:/Users/aptay/Dropbox/adiuvo/mission_critical_course/RTL/synplify_unsafe`, with a 'Browse...' button to its right.
- Description:** A large empty text area.
- Preferred HDL type:** A dropdown menu currently set to 'VHDL'.
- Enable block creation:** An unchecked checkbox.

At the bottom of the dialog, there are four buttons: '< Back', 'Next >', 'Finish', and 'Cancel'. The 'Libero System-on-Chip' logo is visible in the bottom left corner of the dialog area.

Lab 1: Overview and Introduction to Libero

Step 4 – Select a RT3PE3000L device

New project

← **Device selection**
Select a part for your project from the part number list

Selected part: RT3PE3000L-CG484M(1.5V)

Part filter

Family: ProASIC3L Die: All Package: All
Speed: All Core voltage: All Range: All

Reset filters

Search part:

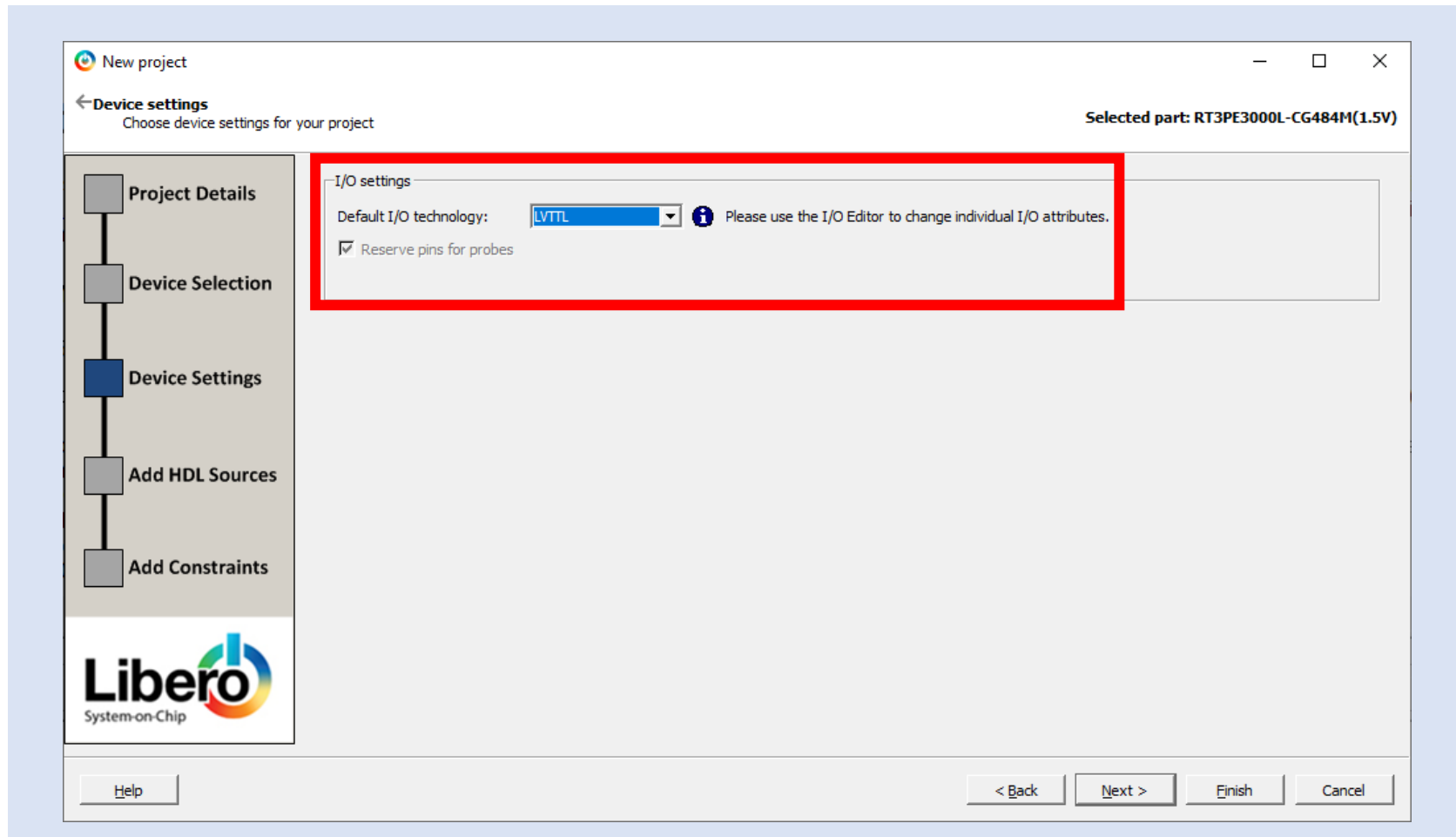
Part Number	VersaTiles (D-Flip)	Maximum User I/O	Differential IO	VersaNet Globals	Integrated PLLs	4608-Bit Block RAM
RT3PE3000L-1CG896M(1.5V)	75264	620	310	18	6	112
RT3PE3000L-1CQ256M(1.2~1.5V)	75264	166	82	18	6	112
RT3PE3000L-1CQ256M(1.2V)	75264	166	82	18	6	112
RT3PE3000L-1CQ256M(1.5V)	75264	166	82	18	6	112
RT3PE3000L-1LG484M(1.2~1.5V)	75264	341	168	18	6	112
RT3PE3000L-1LG484M(1.2V)	75264	341	168	18	6	112
RT3PE3000L-1LG484M(1.5V)	75264	341	168	18	6	112
RT3PE3000L-1LG896M(1.2~1.5V)	75264	620	310	18	6	112
RT3PE3000L-1LG896M(1.2V)	75264	620	310	18	6	112
RT3PE3000L-1LG896M(1.5V)	75264	620	310	18	6	112
RT3PE3000L-CG484M(1.2V)	75264	341	168	18	6	112
RT3PE3000L-CG484M(1.5V)	75264	341	168	18	6	112
RT3PE3000L-CG896M(1.2~1.5V)	75264	620	310	18	6	112

Libero System-on-Chip

Help < Back Next > Finish Cancel

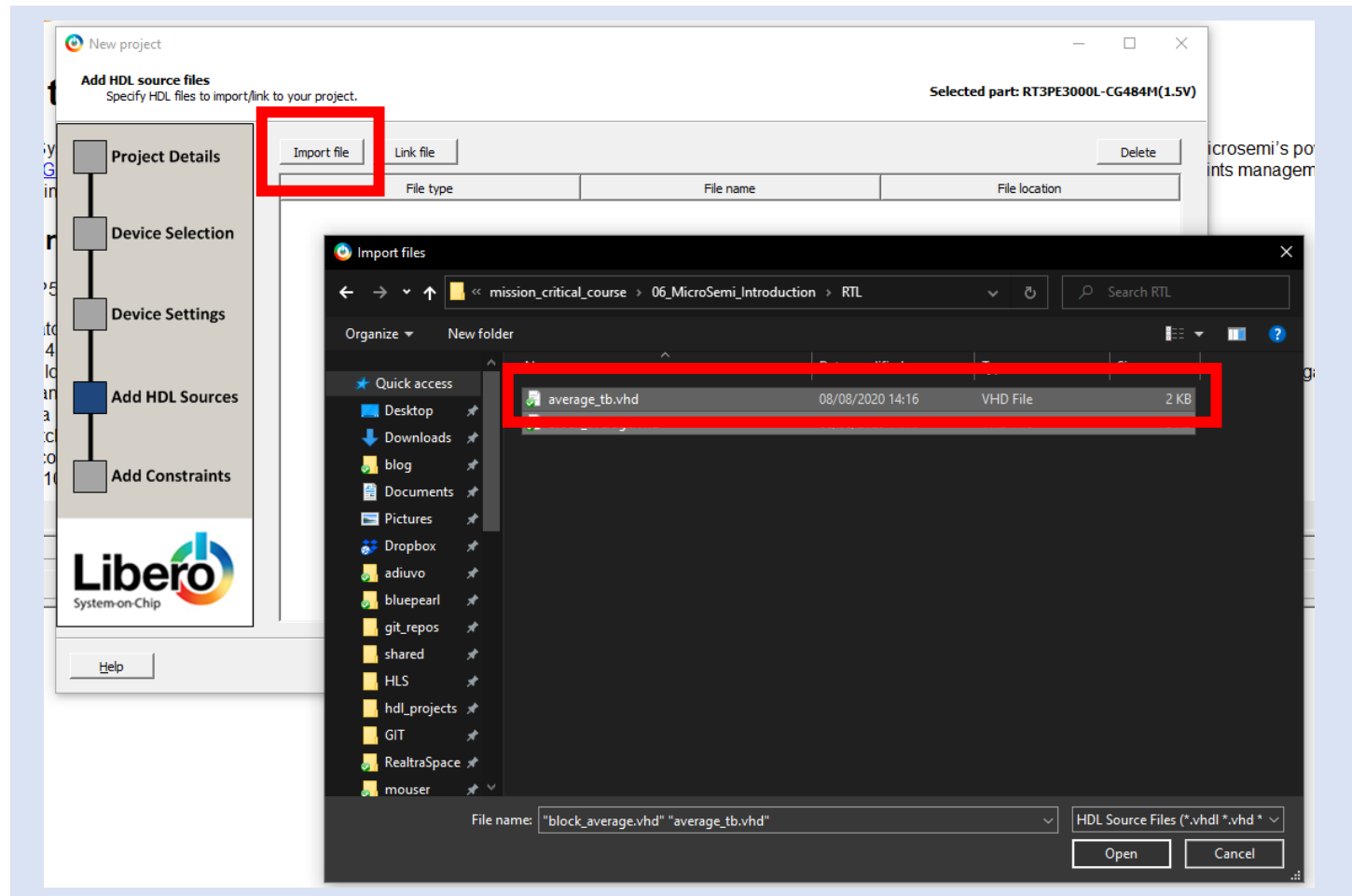
Lab 1: Overview and Introduction to Libero

Step 5 – Select LVTTTL as the Default I/O Technology



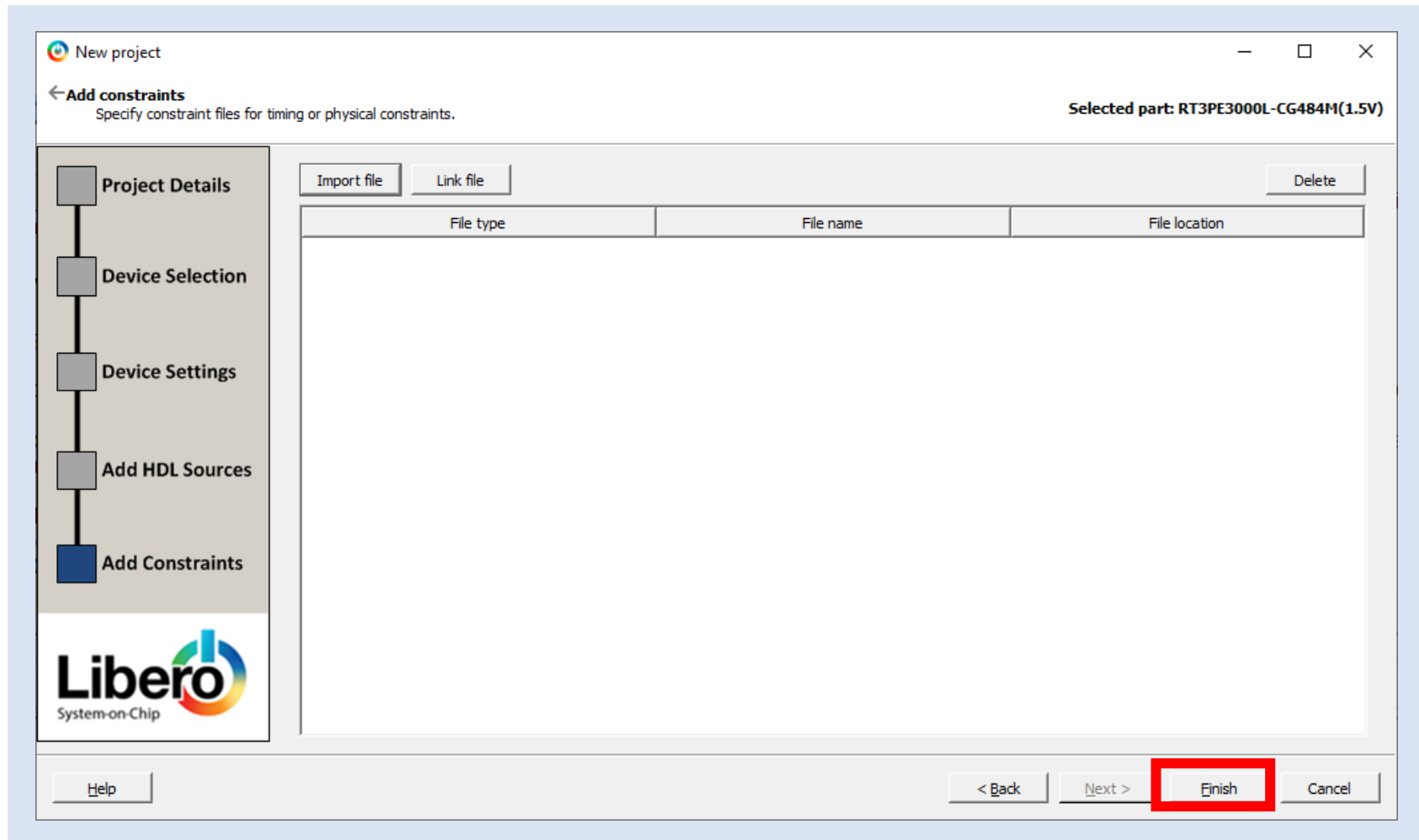
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Step 6 – Select import file and import the block_average.vhd example file provided



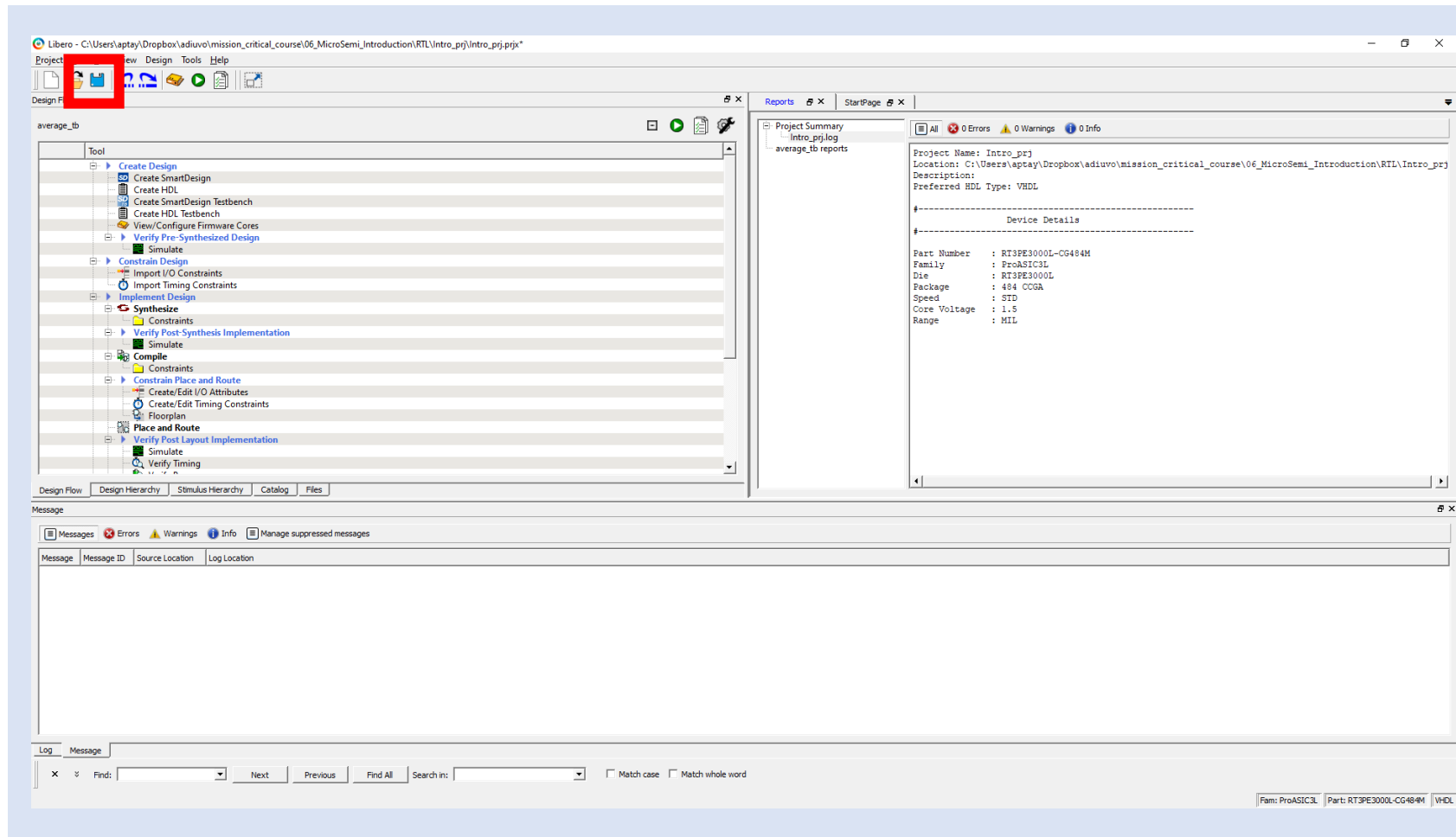
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Step 7 – Leave the constraint empty and select finish



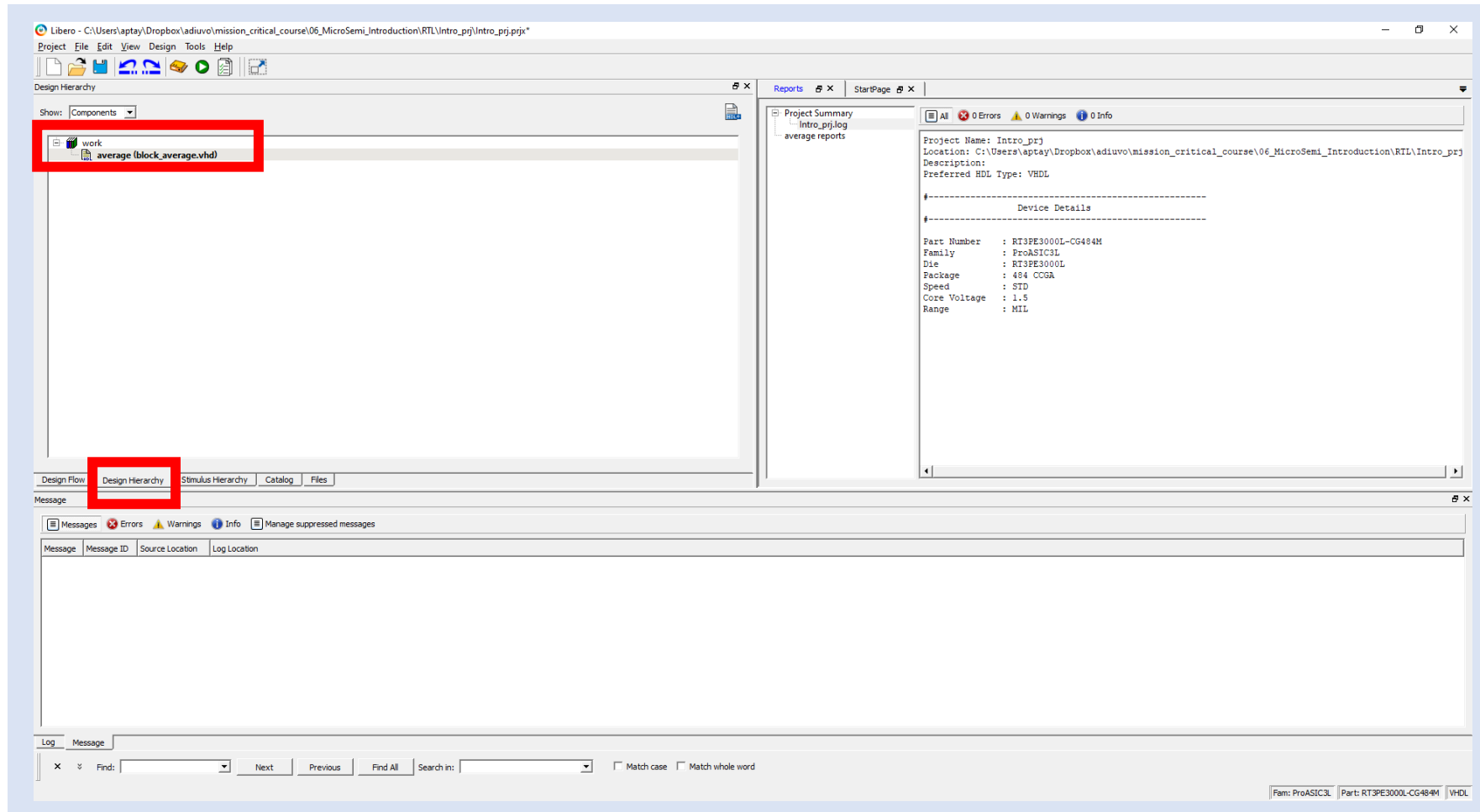
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Step 8 – This will open the project in Libero – Click on Save



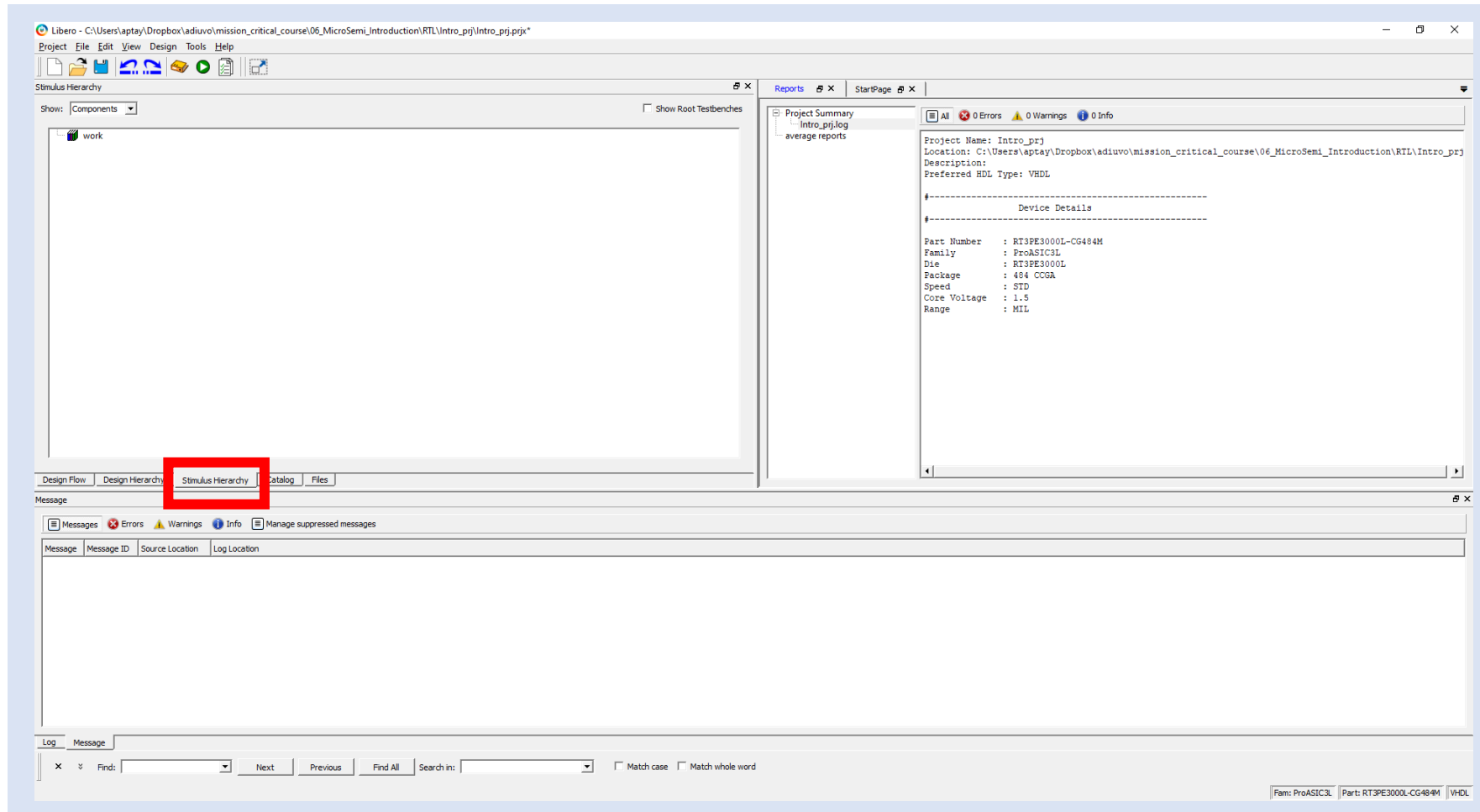
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Step 9 – Select Design Hierarchy – You will should see the RTL file



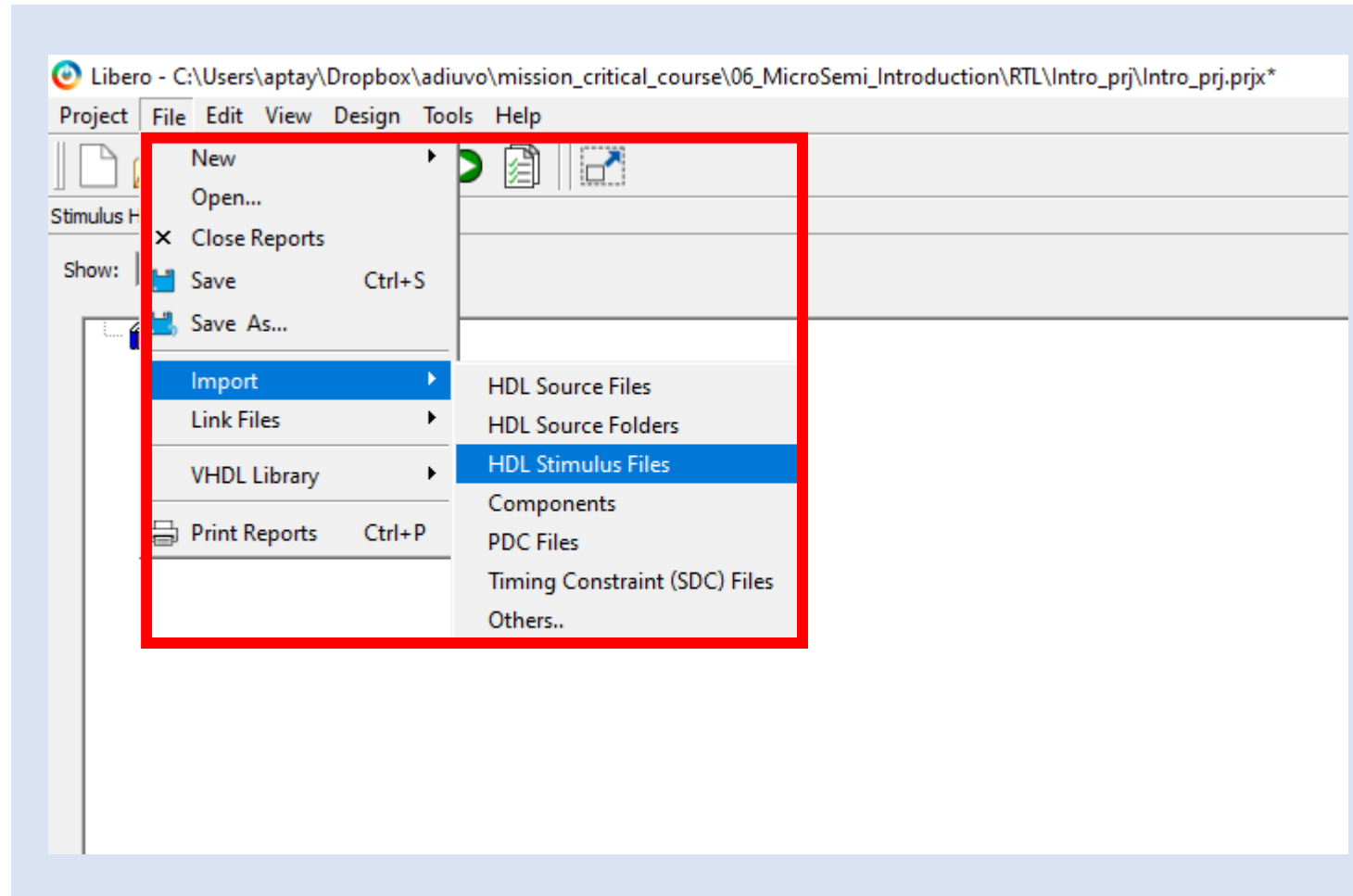
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Step 10 – Select the stimulus hierarchy tab – notice it is empty



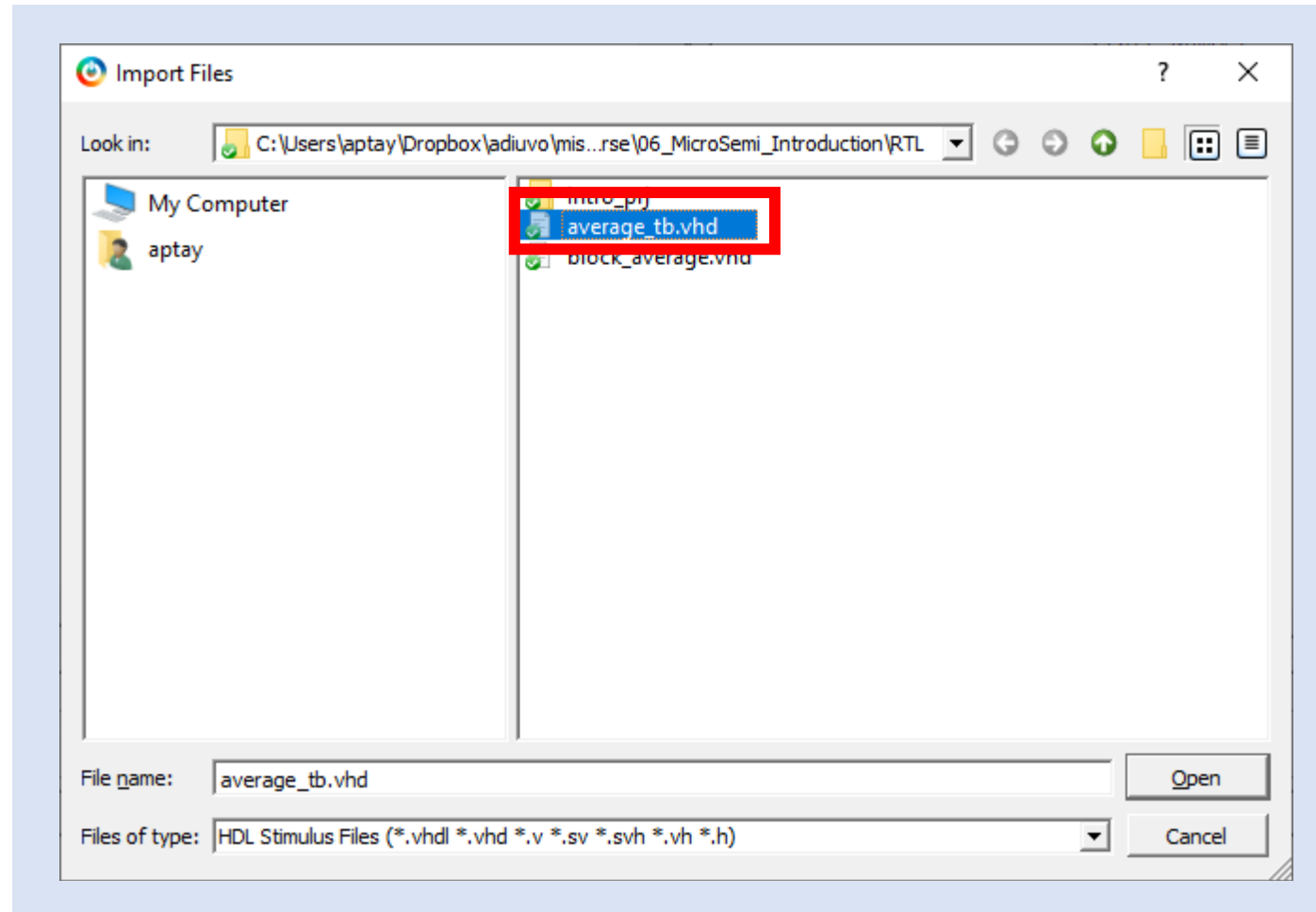
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Step 11 – Select File -> Import -> HDL Stimulus Files



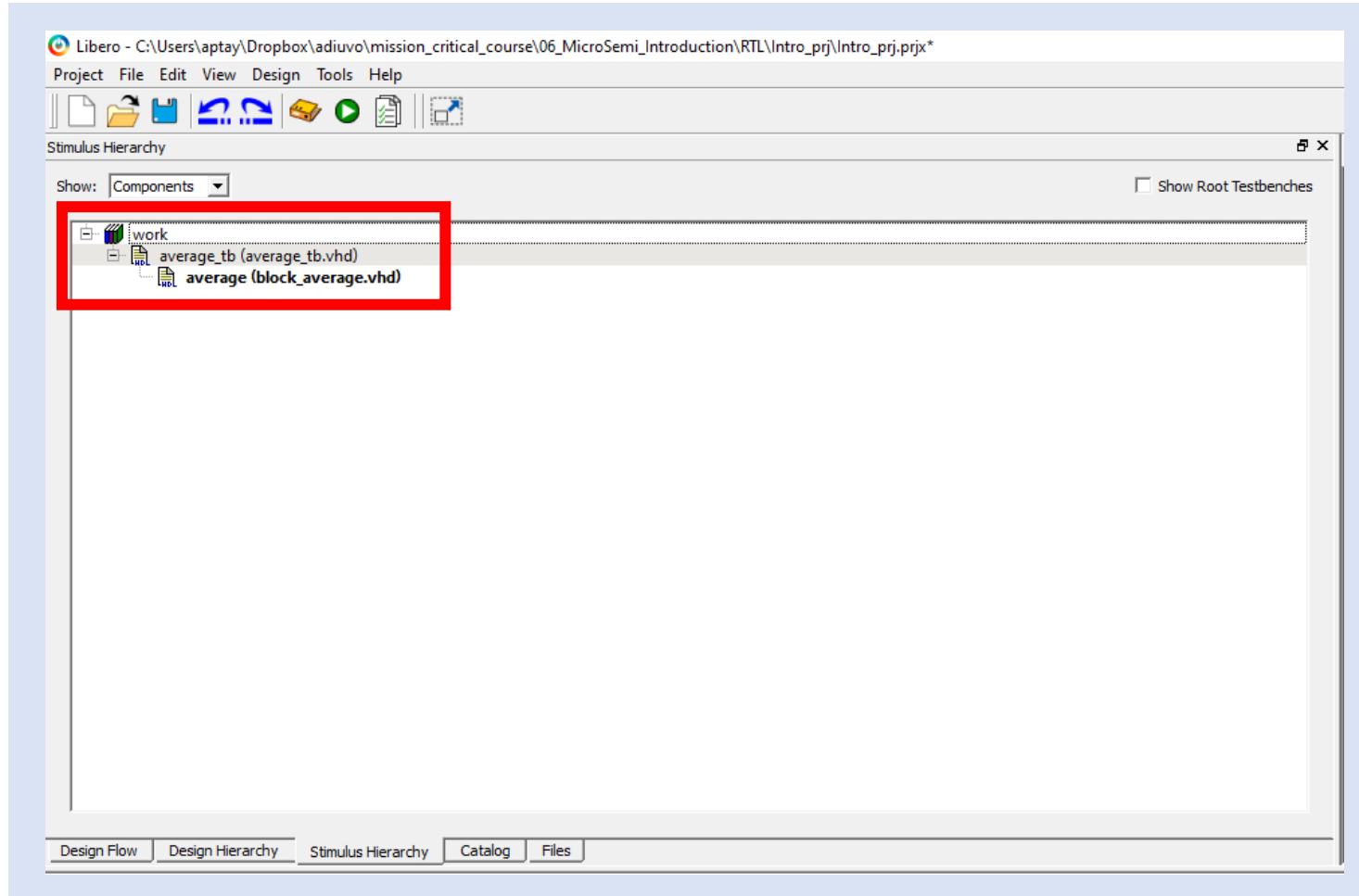
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Step 12 – Select File, the average_tb.vhd file



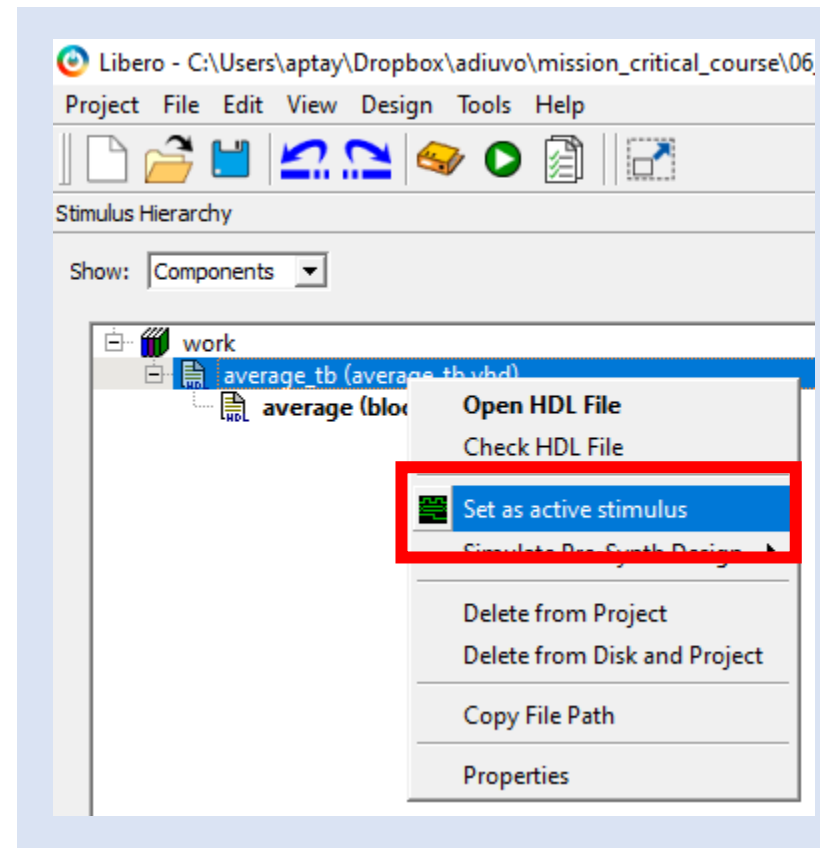
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Step 13 – The file will show in the Stimulus Hierarchy, expanding the average_tb file should show the block_average RTL file.



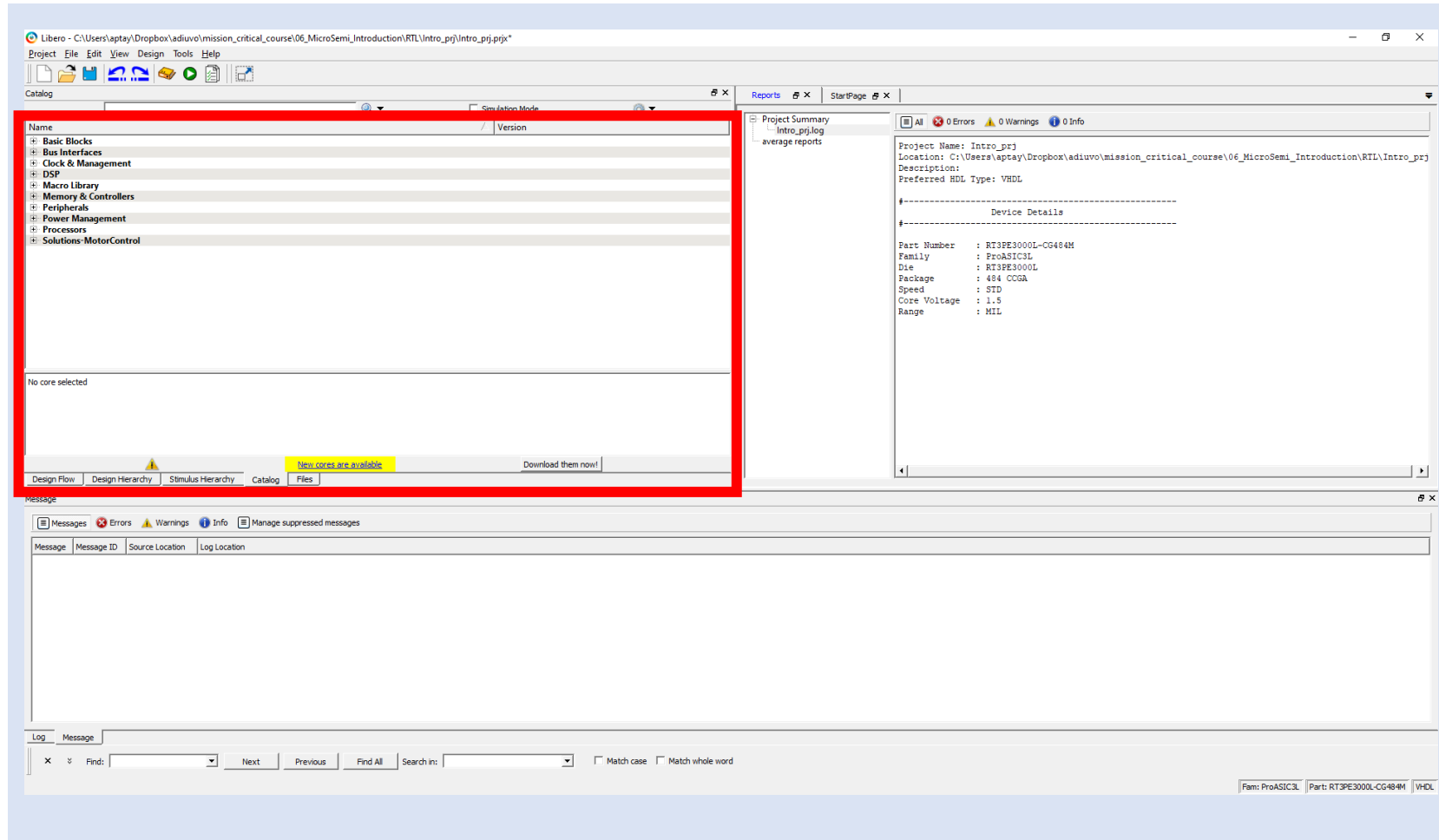
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Step 15 – Right click on the test bench and select – Set as Active Stimulus



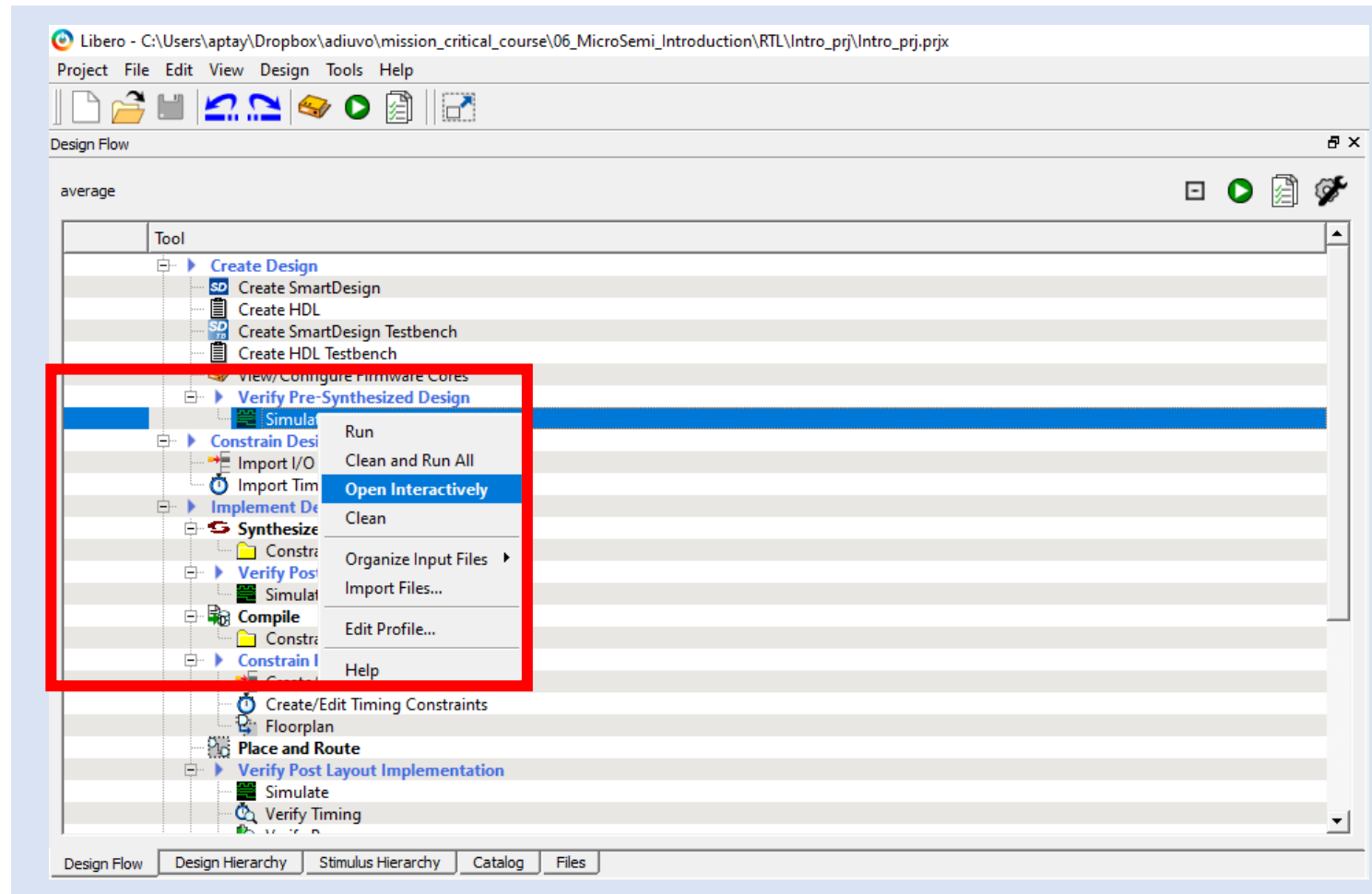
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Step 16 – Click on Catalog and you will see the IP which can be generated for inclusion in the design



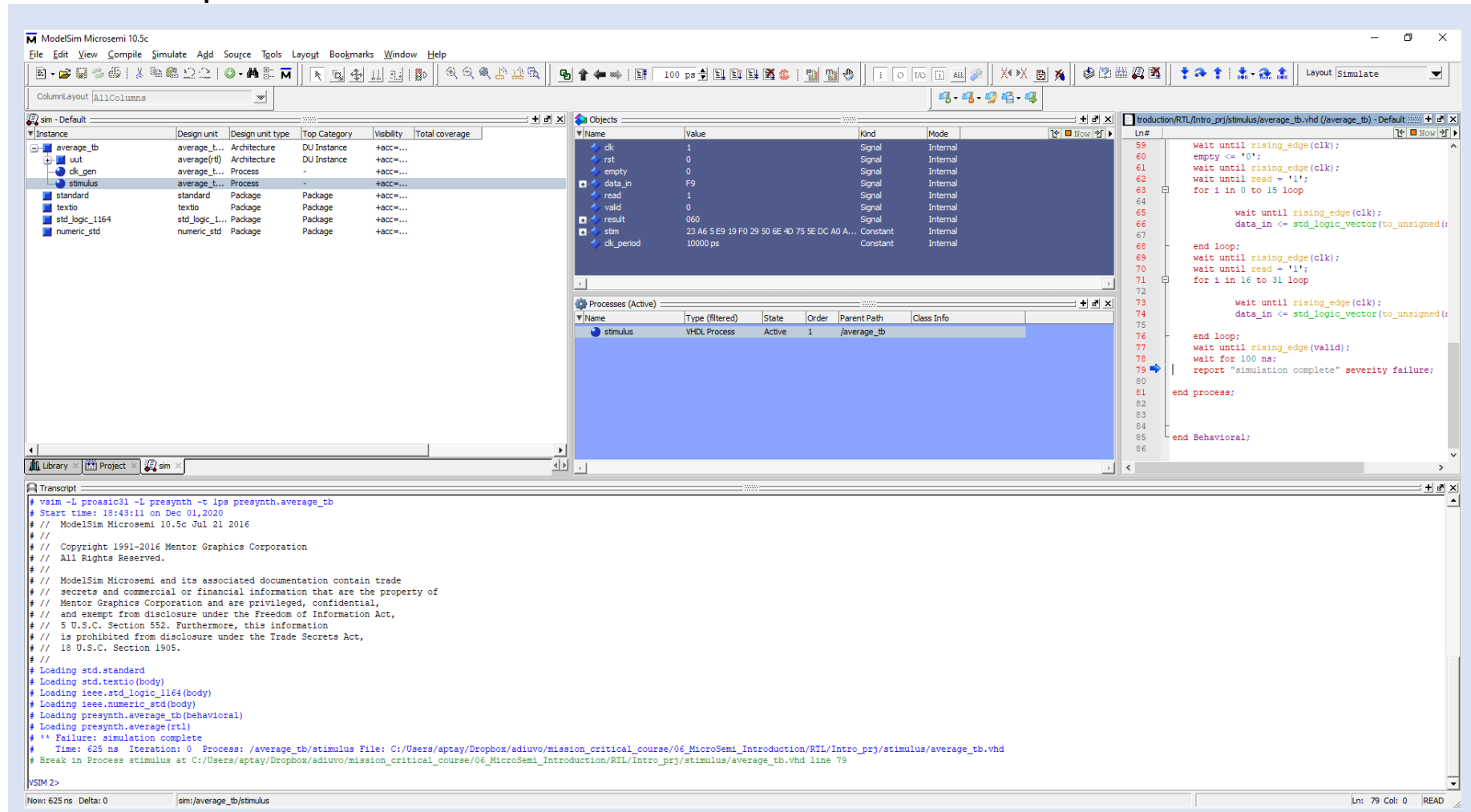
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Step 17 – Back on the Design Flow Tab, right click on simulate and select Open Interactively



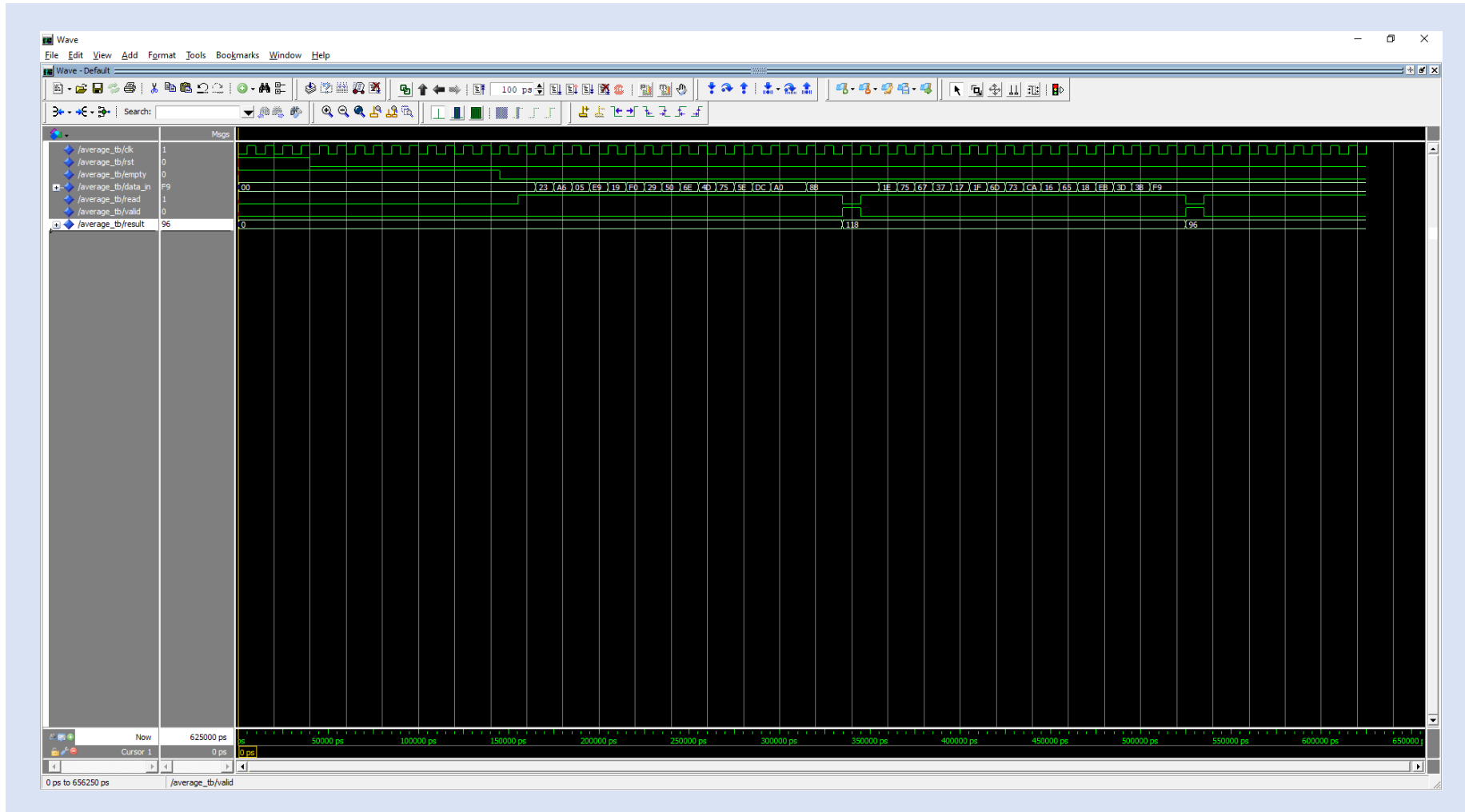
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Step 18 – This will open ModelSim and run the stimulus file.



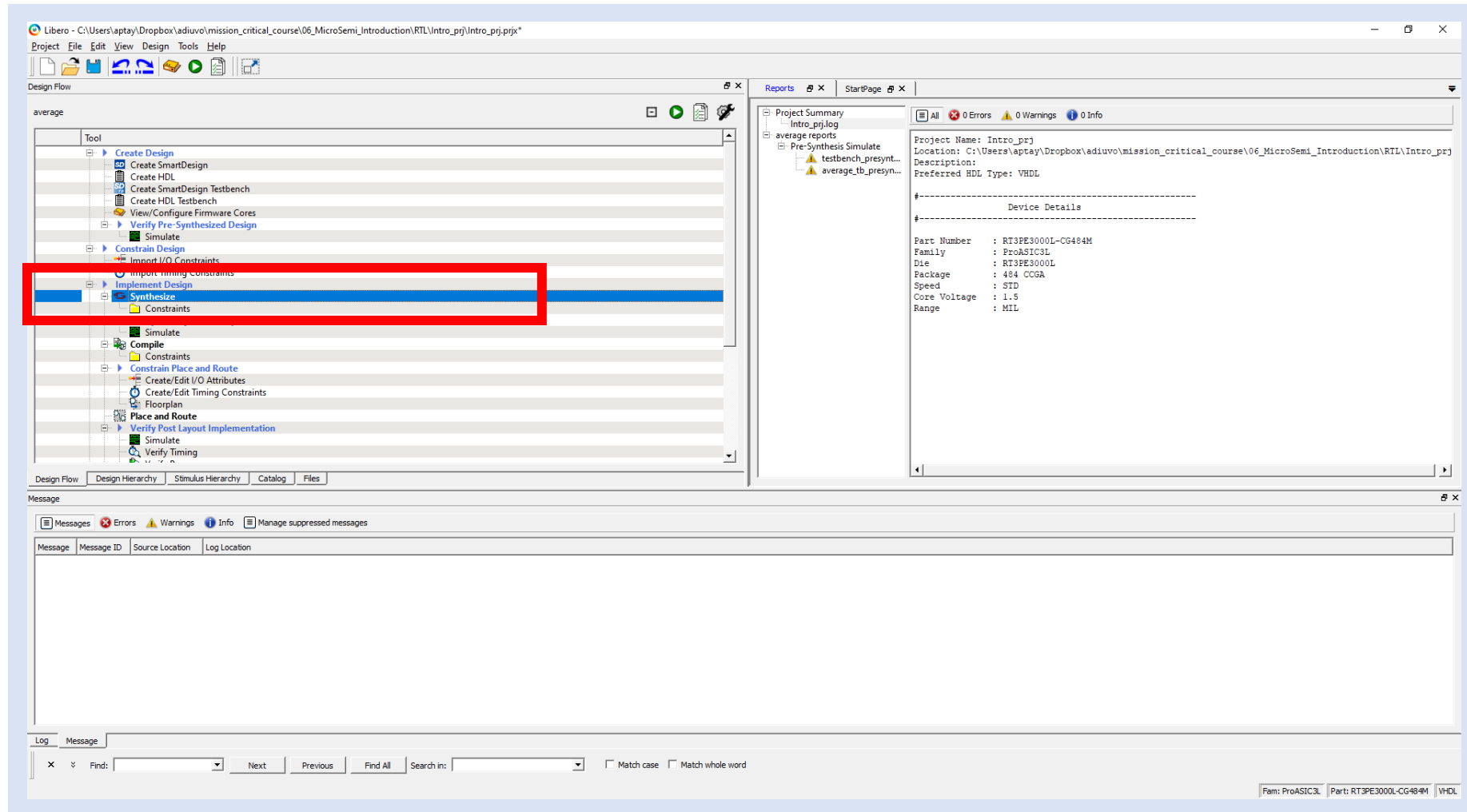
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Step 19 – Click on the waveform, you will see the average block input and resultant output



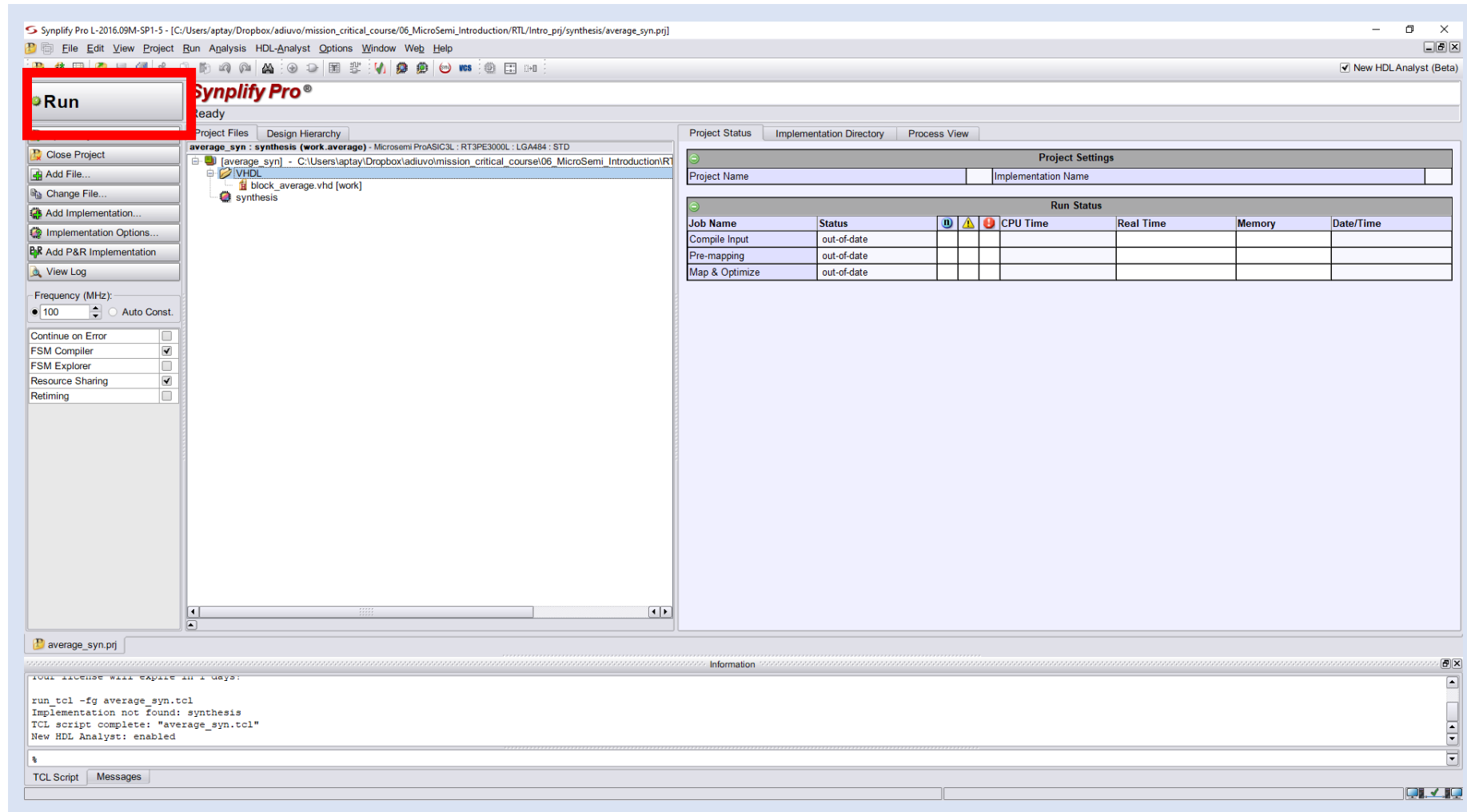
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Step 20 – Close ModelSim and Return to Libero Project Management Flow – Select Synthesize and Open Interactively



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Step 21 – Once Synplify Pro opens, expand the VHDL folder and check the block average file is present – Click on run this will synthesis the design



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Step 22 – Once the Synthesis is completed check there are no errors and open the RTL schematic

The screenshot shows the Synplify Pro L-2016.09M-SP1-5 interface. The top status bar indicates "Done: 0 errors, 8 warnings, 18 notes". The left sidebar shows the "Run" tab with a list of project files, including "average_syn : synthesis (work.average)". The main window displays the "Project Settings" tab, which includes a table of project settings, a "Run Status" table, an "Area Summary" table, a "Timing Summary" table, and an "Optimizations Summary" table.

Project Settings

Project Name	average_syn	Device Name	synthesis: Microsemi ProASIC3L : RT3PE3000L
Implementation Name	synthesis	Top Module	work.average
Retiming	0	Resource Sharing	1
Fanout Guide	24	Disable I/O Insertion	0
Disable Sequential Optimizations	0	FSM Compiler	1

Run Status

Job Name	Status	CPU Time	Real Time	Memory	Date/Time
Compile Input (compiler)	Complete	0	00m:00s	-	01/12/2020 18:59:08
Pre-mapping (premap)	Complete	0	00m:00s	110MB	01/12/2020 18:59:08
Map & Optimize (fpga_mapper)	Complete	0	00m:01s	112MB	01/12/2020 18:59:11

Area Summary

Core Cells	IO Cells
86	25
Block RAMs (v_ram)	0

Timing Summary

Clock Name (clock_name)	Req Freq (req_freq)	Est Freq (est_freq)	Slack (slack)
averageclk	100.0 MHz	103.5 MHz	0.335

Optimizations Summary

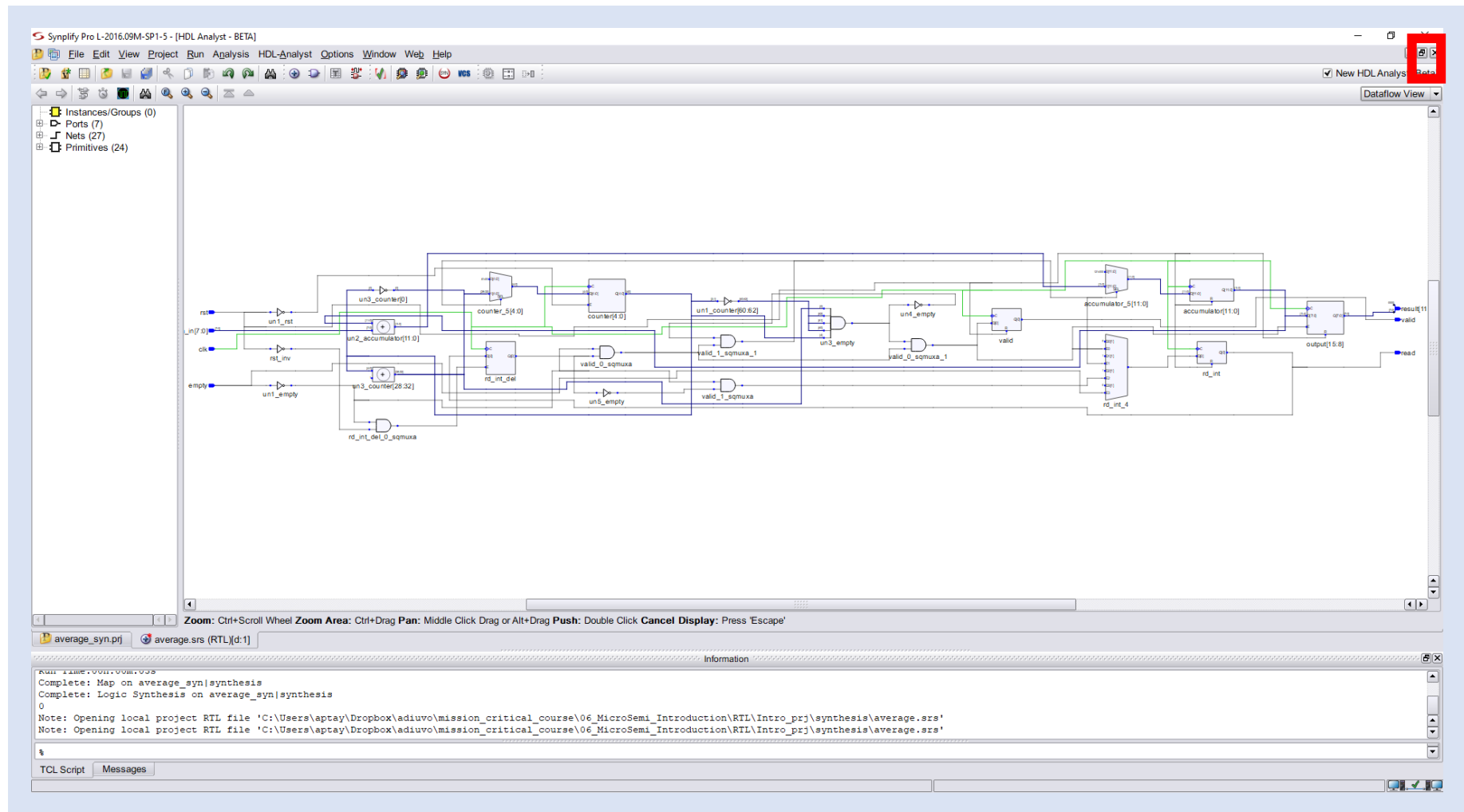
Combined Clock Conversion	1 / 0
more	

The bottom status bar shows the "TCL Script" and "Messages" tabs. The "Messages" tab displays the following information:

```
Return Code: 1
Run Time:00h:00m:03s
Complete: Map on average_syn|synthesis
Complete: Logic Synthesis on average_syn|synthesis
0
Note: Opening local project RTL file 'C:\Users\aptay\Dropbox\adiuvo\mission_critical_course\06_MicroSemi_Introduction\RTL\Intro_prj\synthesis\average.srs'
```

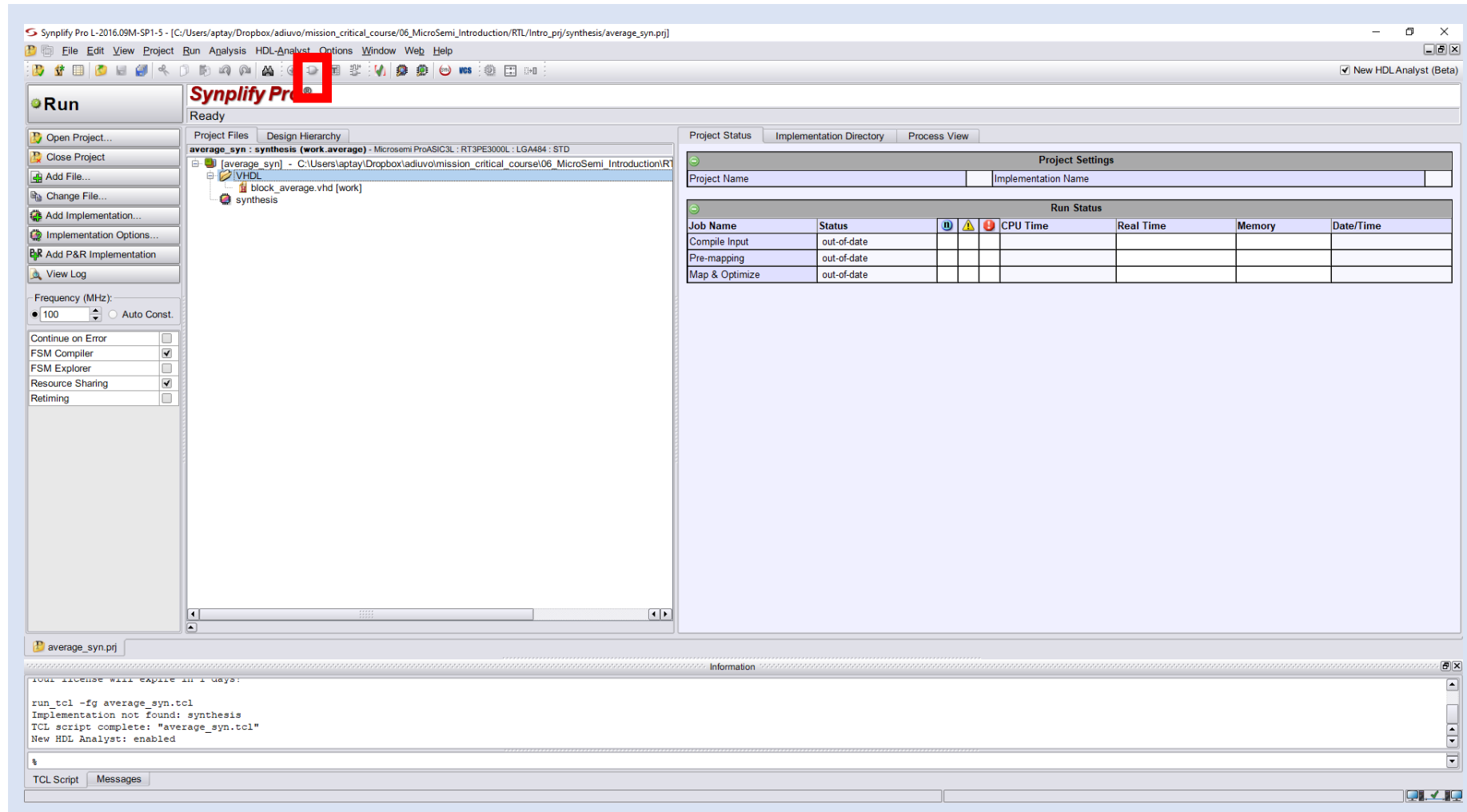
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Step 23 – The RTL View will show the RTL netlist of your design – This can be useful later when we are looking at implementing high reliability solutions. Close the RTL View.



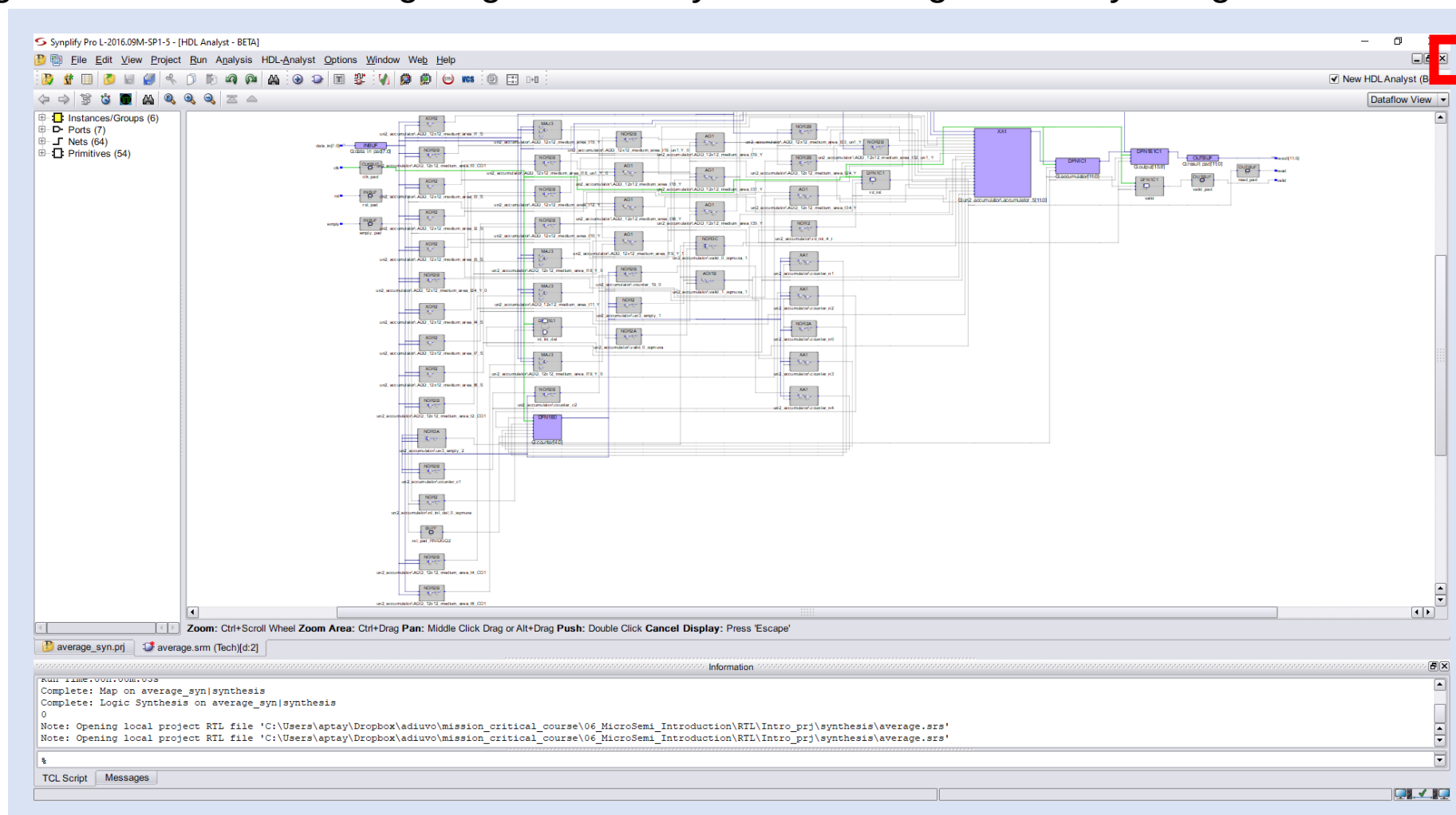
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Step 24 – Open the Technology View.



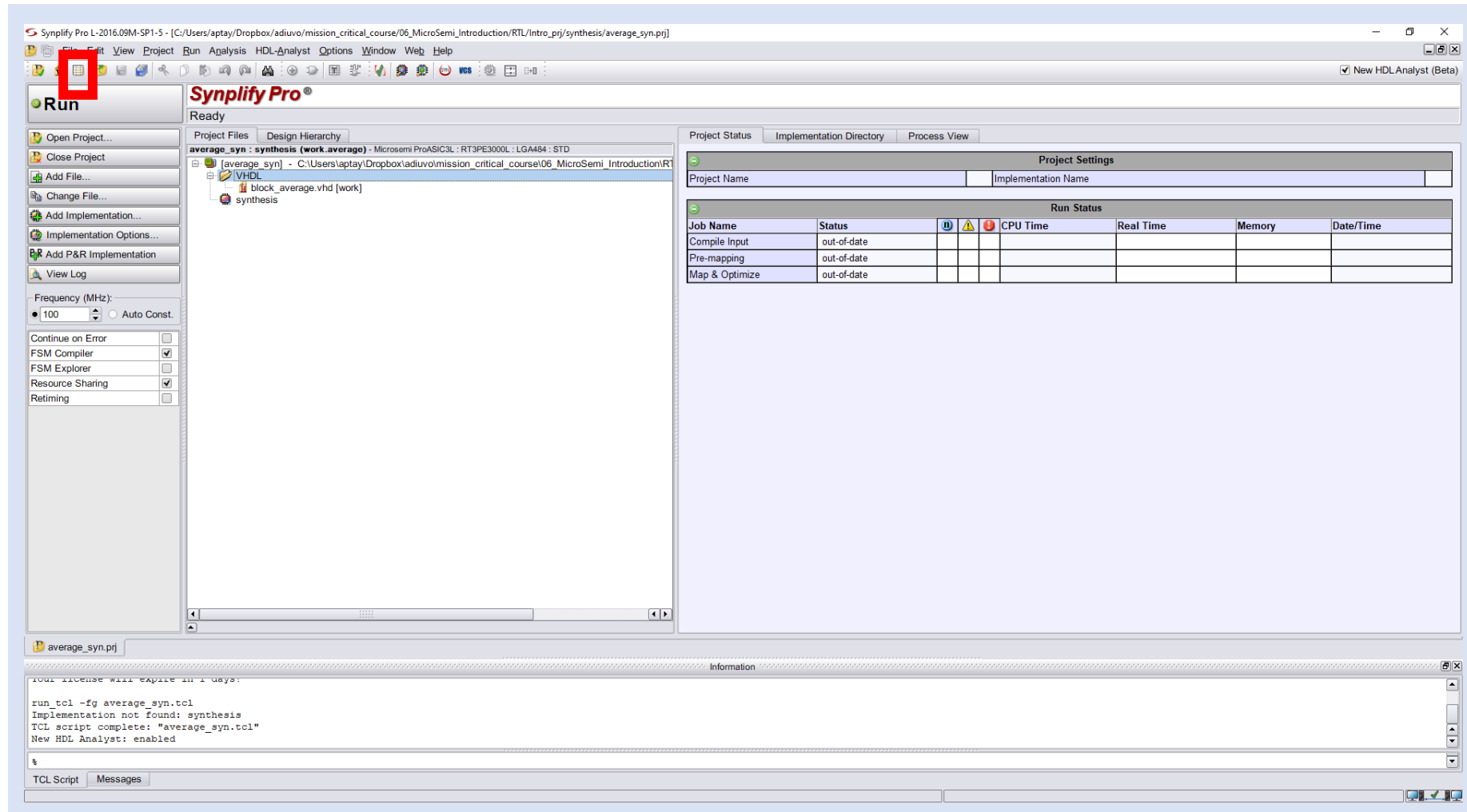
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Step 25 – The Technology View shows the implemented logic resources in the target device. This view is again useful when investigating issues in synthesis for high reliability designs



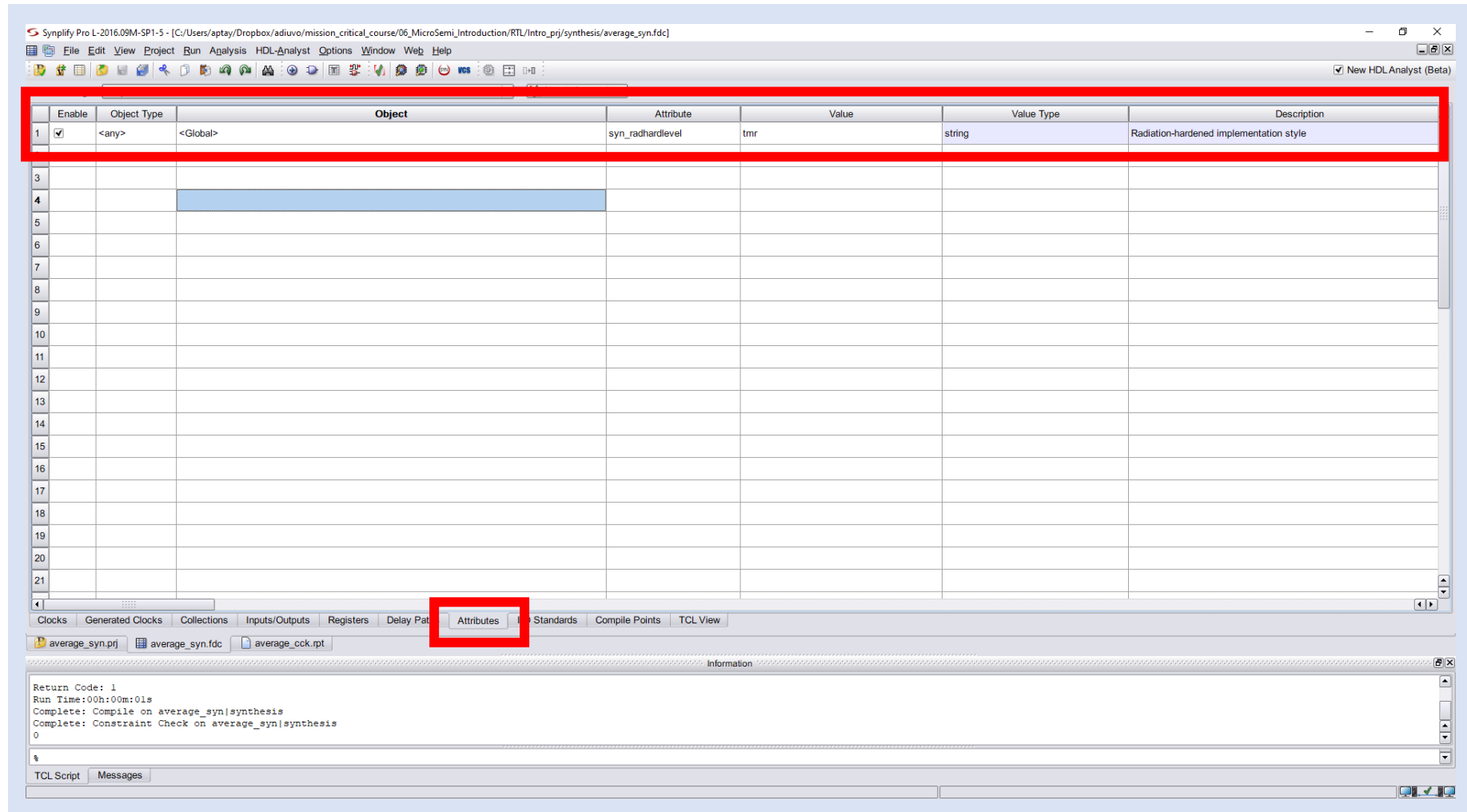
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Step 26 – If we need to add synthesis constraints e.g. design TMR we can do this via the SCOPE editor



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Step 27 – Click on the Attributes Tab and select a Syn_RADHARDLEVEL of TMR



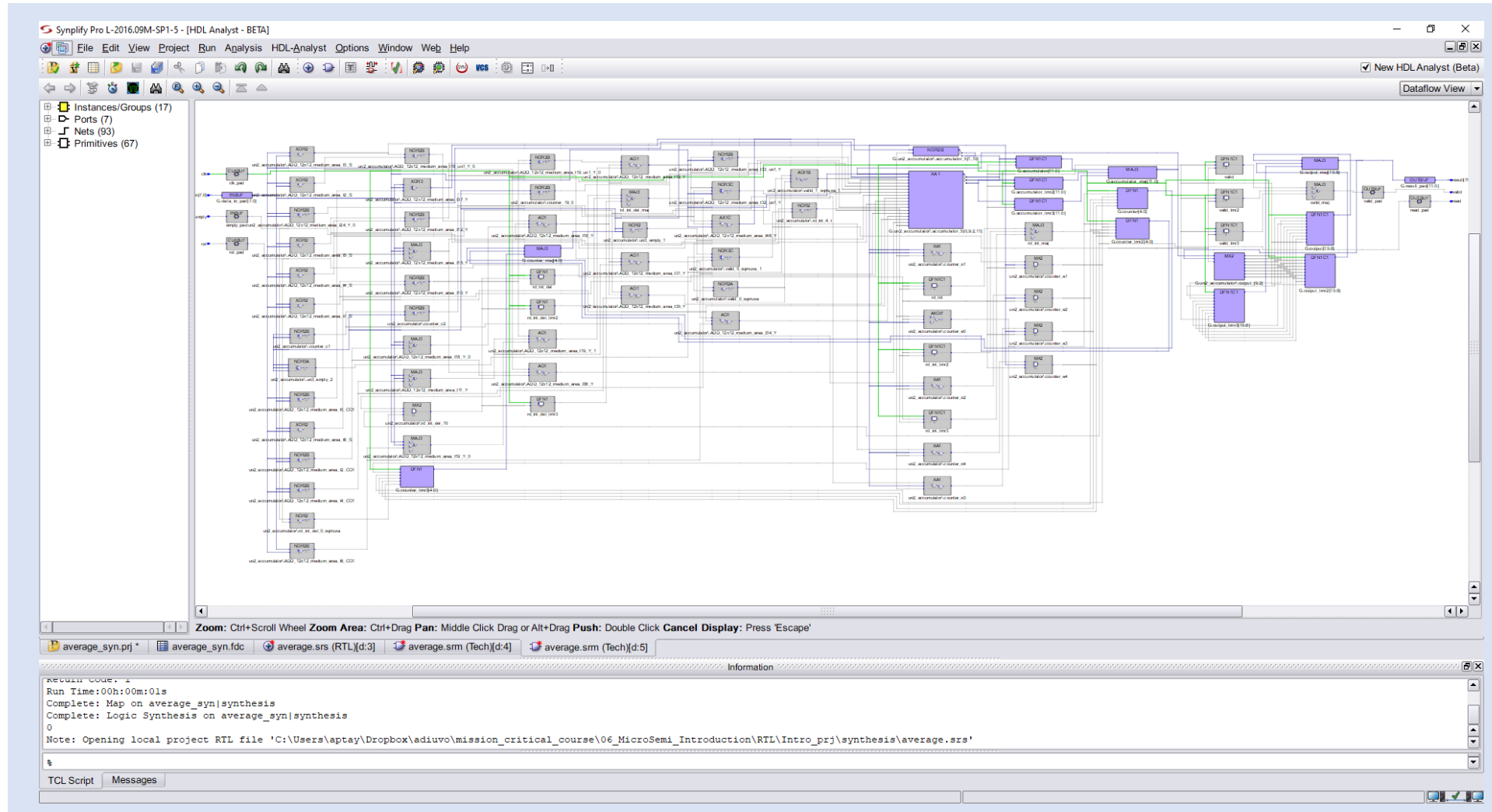
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Step 28 – Save the file and when prompted add in the saved file to the project



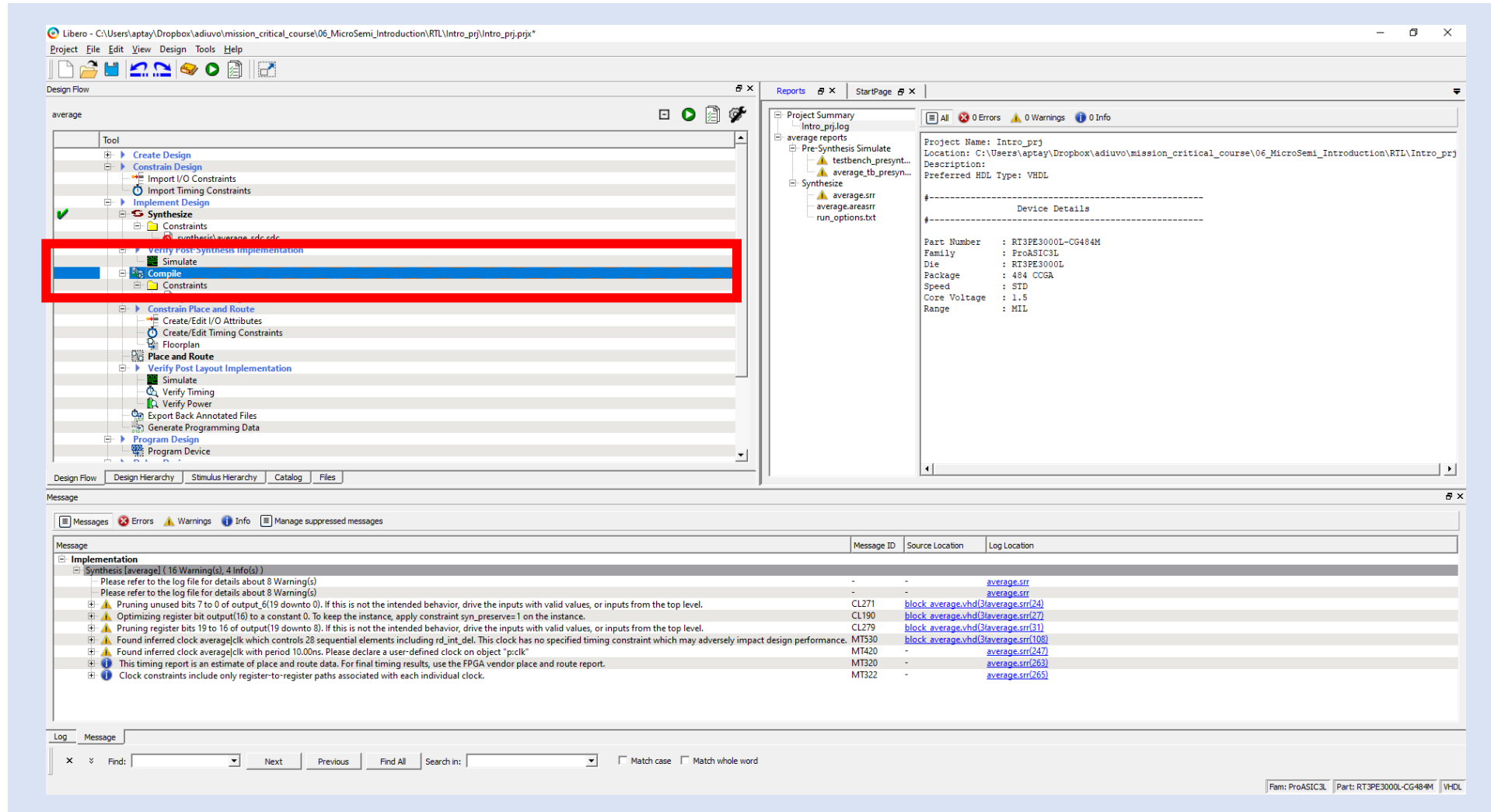
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Step 29 – Re-Run the Synthesis and examine the Technology view this will present TMR implementation. Explore the design then close Synplify and return to Libero



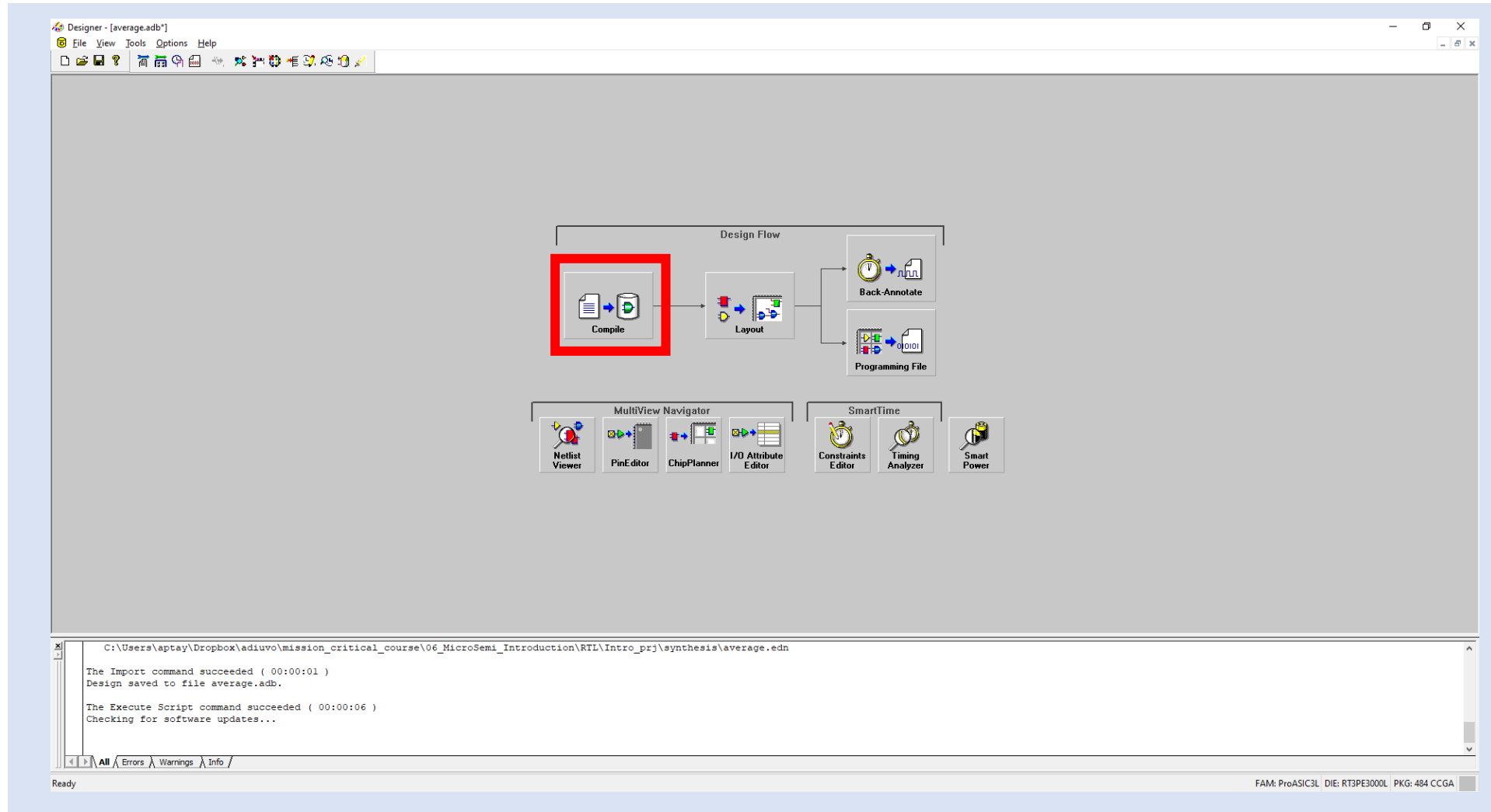
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Step 30 – Back in Libero select the Compile and again right click and open interactively



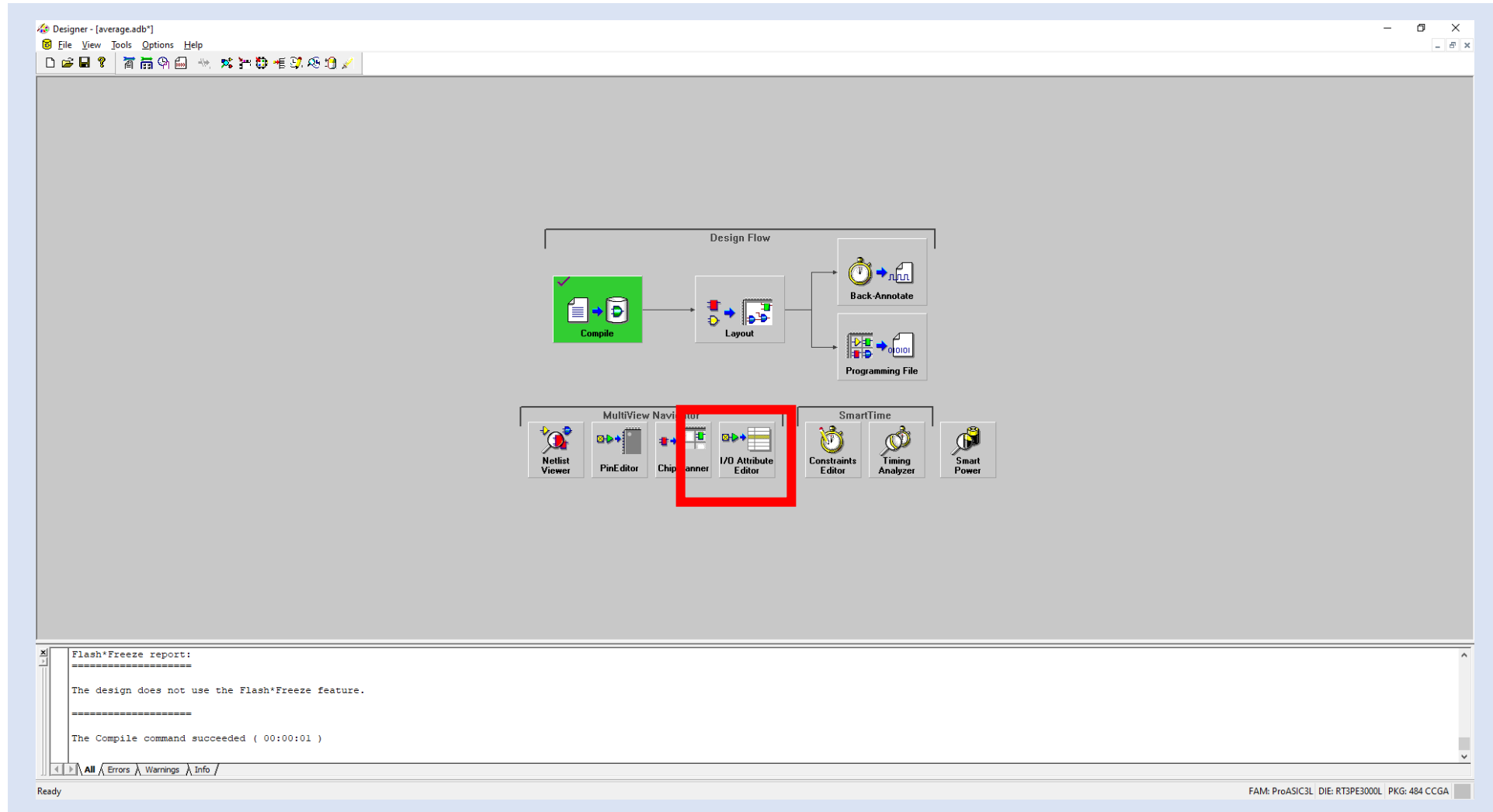
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Step 31 – In Designer click on Compile - If a compile option dialog appears leave unchanged and click OK



Lab 1: Overview and Introduction to Libero

Step 32 – Click on I/O Attribute Editor



Lab 1: Overview and Introduction to Libero

Step 32 – Click on I/O Attribute Editor – Make sure clock is on D5

The screenshot shows the I/O Attribute Editor window. The 'Edit' button in the top-left toolbar is highlighted with a red box. The main table lists various ports and their attributes. The 'clk' port is highlighted in row 1.

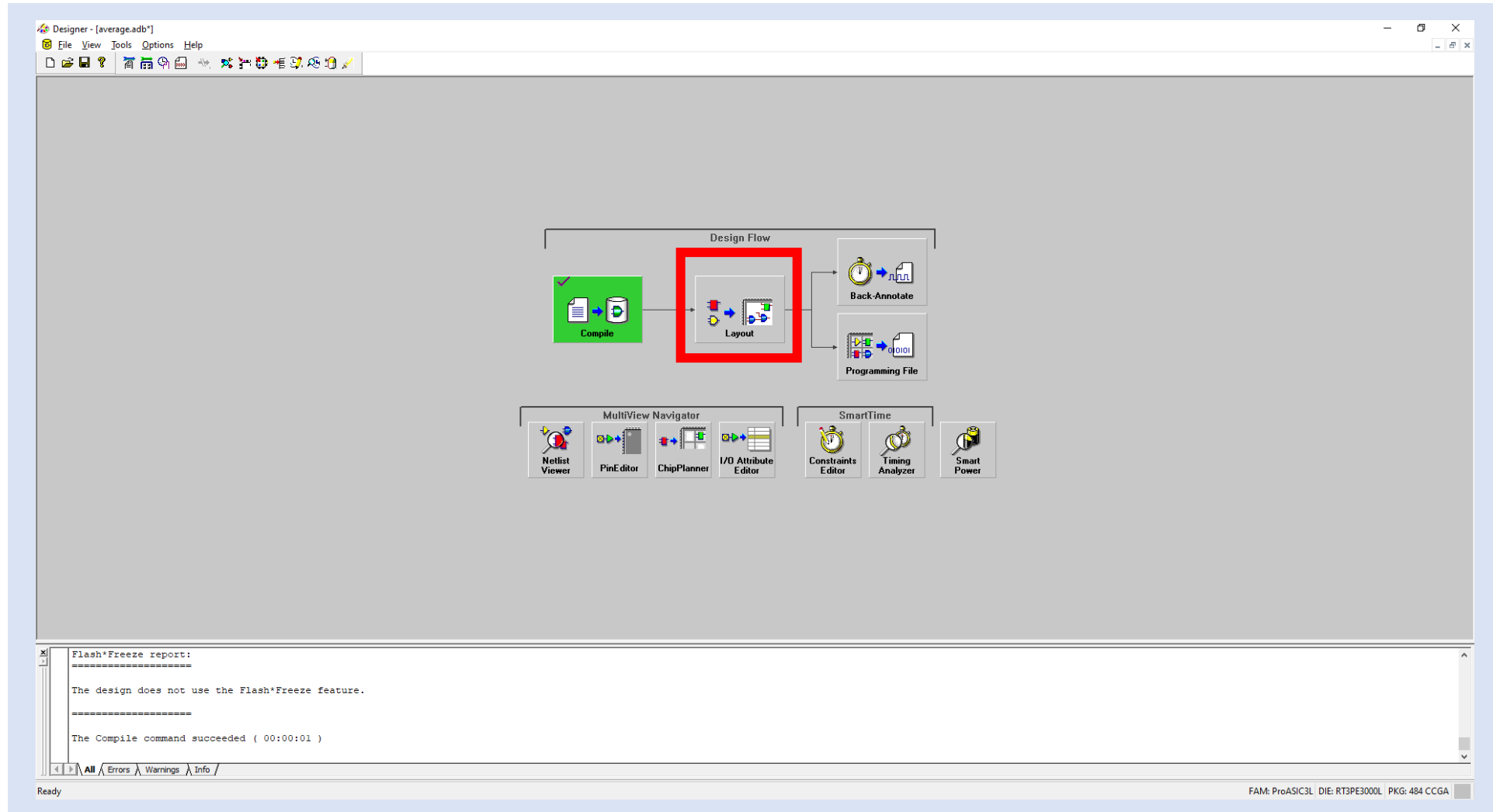
	Port Name/	Group	Macro Cell	Pin Number	Locked	Bank Name	I/O Standard	Output Drive (mA)	Slew	Resistor Pull	Schmitt Trigger	Input Delay	Skew	Output Load (pF)	Use I/O Reg	Hot Swappable
1	clk		ADLIB.CLKB...	D5	☒	Bank 0	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
2	data_in[0]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
3	data_in[1]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
4	data_in[2]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
5	data_in[3]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
6	data_in[4]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
7	data_in[5]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
8	data_in[6]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
9	data_in[7]		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
10	empty		ADLIB.INBUF	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
11	read		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
12	result[0]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
13	result[1]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
14	result[2]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
15	result[3]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
16	result[4]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
17	result[5]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
18	result[6]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
19	result[7]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
20	result[8]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
21	result[9]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
22	result[10]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
23	result[11]		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒
24	rst		ADLIB.CLKB...	Unassigned	--	--	LVTTTL	--	--	None	☐	Off	--	--	☐	☒
25	valid		ADLIB.OUTB...	Unassigned	--	--	LVTTTL	12	High	None	--	--	☐	5	☐	☒

Ports View

row 1, col 4 FAM: ProASIC3L DIE: RT3PE3000L PACKAGE: 484 CCGA

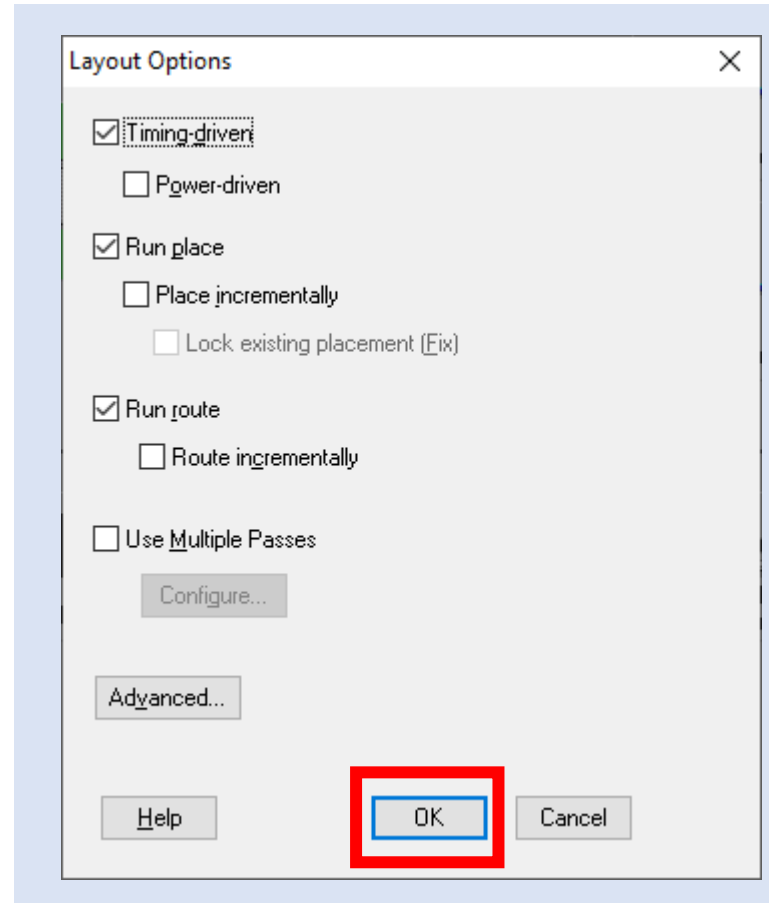
Lab 1: Overview and Introduction to Libero

Step 32 – Click on I/O Attribute Editor



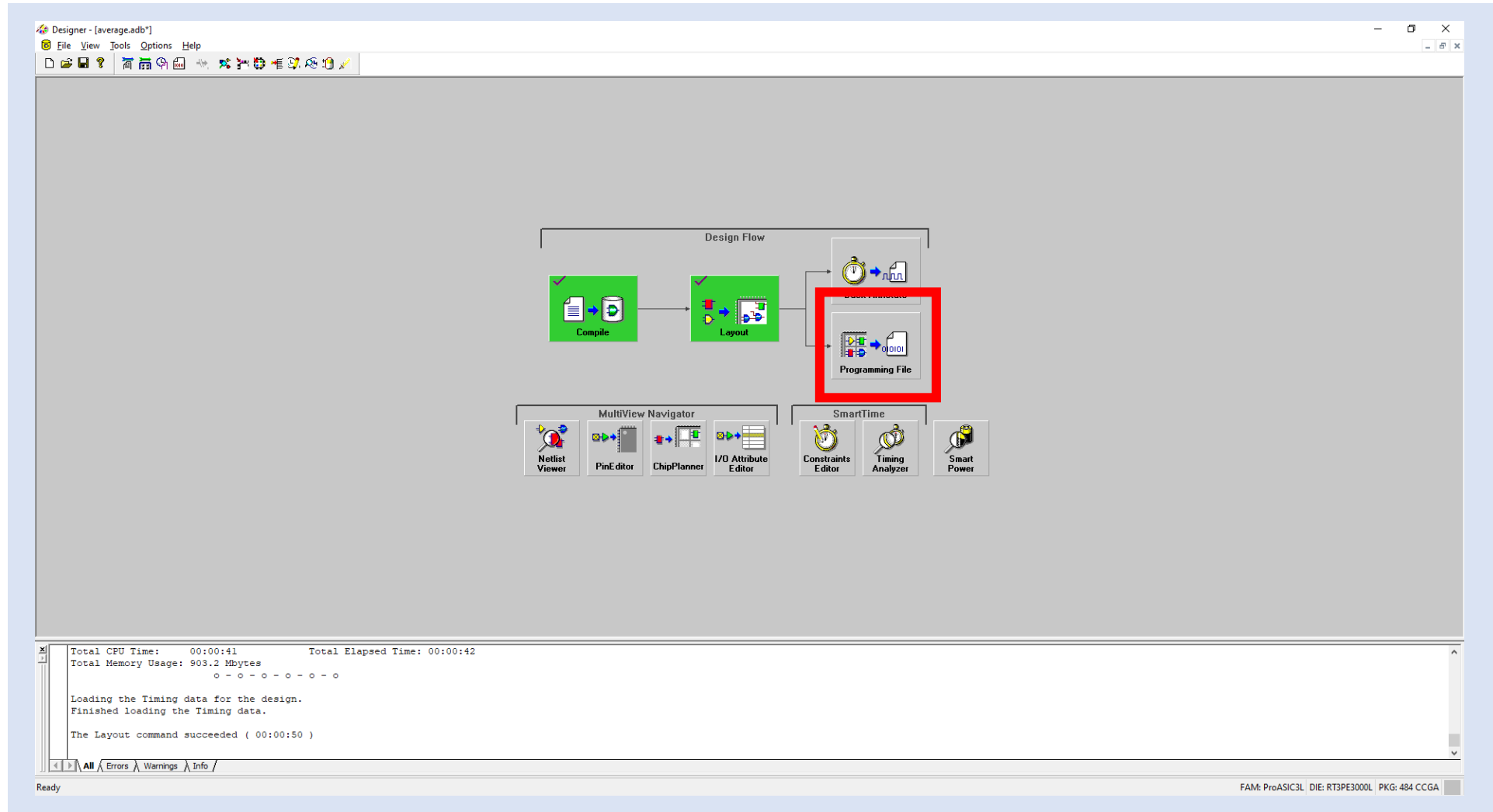
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Step 32 – Click on I/O Attribute Editor



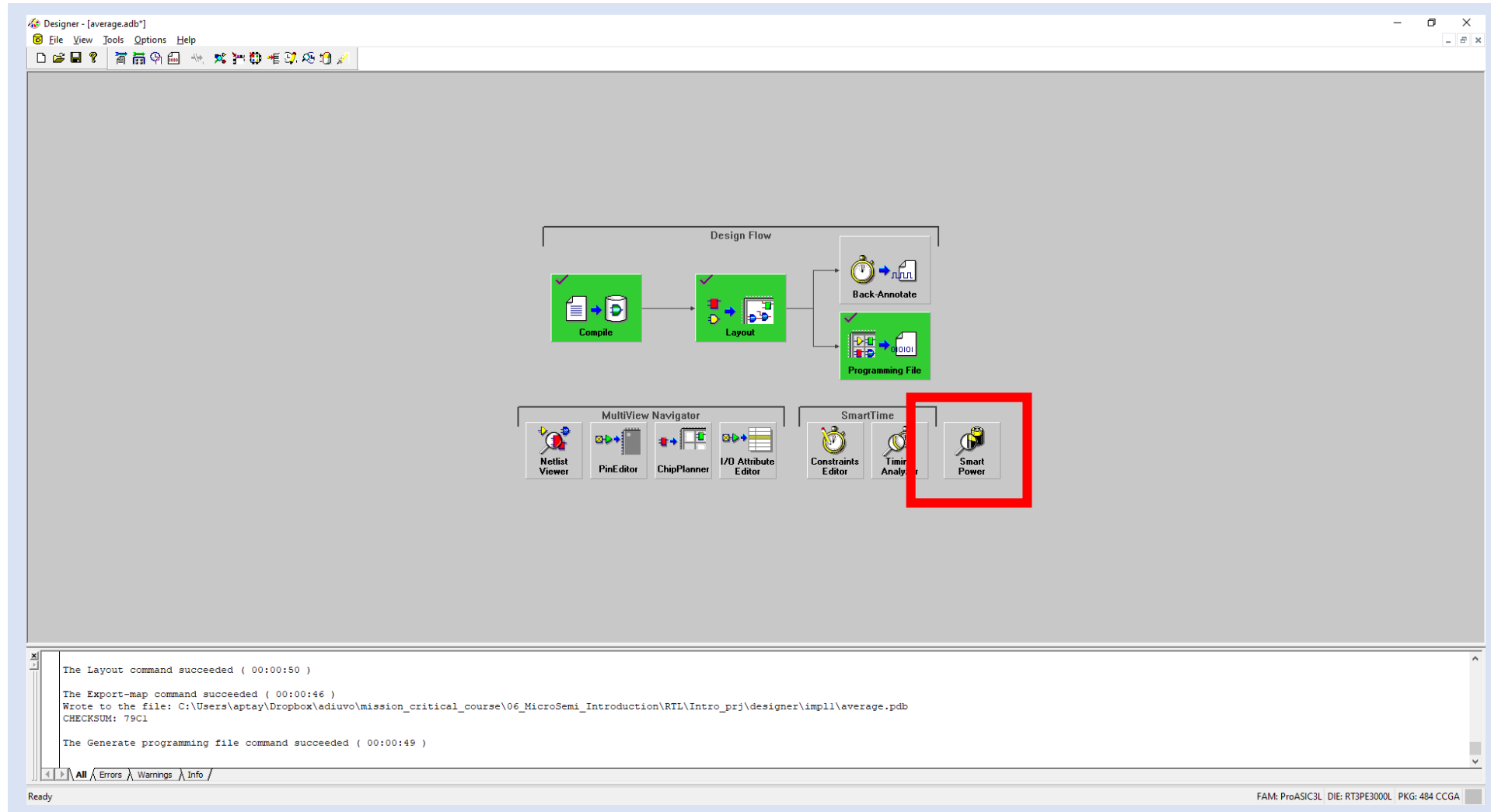
Lab 1: Overview and Introduction to Libero

Step 32 – Click on I/O Attribute Editor



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